MILITARY SPECIFICATION

TERMINAL, VOICE FREQUENCY CARRIER TELEGRAPH (VFCT), AN/UCC-1( )(V)

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers shipboard multichannel voice frequency carrier telegraph (VFCT) terminal, AN/UCC-1( )(V), hereinafter referred to as the VFCT terminal, data communications equipment (DCE) for use in long haul high frequency (HF) radio circuits.

1.2 Classification. The VFCT terminal shall be of the functional mode, in-band frequency diversity, and resident channel capabilities specified in 1.2.1, 1.2.2 and 1.2.3 (see 6.2).

1.2.1 Functional mode capability. The functional mode capabilities of the VFCT terminal shall be as classified in a and b:

a. Modulation and demodulation (VFCT terminal capable of both modulation and demodulation)
b. Demodulation (VFCT terminal capable of demodulation)

The functional mode classification specified in a and b shall not preclude a VFCT terminal which is classified as modulation and demodulation functional mode capable from also being classified as demodulation functional mode capable if the VFCT terminal can also function in a demodulation only application.

1.2.2 In-band frequency diversity capability. The modulation function and the demodulation function of the VFCT terminal shall be each classified by in-band frequency diversity capability.

1.2.2.1 Modulation function in-band frequency diversity capability. The modulation function of the modulation and demodulation functional mode capable VFCT terminal shall be as classified in a and b:

a. Untwinned modulation channels (VFCT terminal capable of untwinned modulation channel operation (see 3.4.2.3.1))
b. Untwinned and twinned modulation channels (VFCT terminal capable of, in turn, untwinned modulation channel operation (see 3.4.2.3.1) and twinned modulation channel operation (see 3.4.2.3.2))

The classification specified in a and b shall not preclude a VFCT terminal whose modulation function is capable of providing untwinned and twinned modulation channels from also being classified as providing untwinned modulation channel operation if the VFCT terminal can so function in an untwinned modulation only application.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5805

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1.2.2.2 Demodulation function in-band frequency diversity capability. The demodulation function of the modulation and demodulation functional mode capable VFCT terminal and the demodulation functional mode capable VFCT terminal shall be as classified in a and b:

a. Untwinned demodulation channels (VFCT terminal capable of, in turn, nondiversity (see 3.4.3.2.1) and untwinned audio input diversity (see 3.4.3.2.3) operation).

b. Untwinned and twinned demodulation channels (VFCT terminal capable of, in turn, nondiversity (see 3.4.3.2.1), untwinned audio input diversity (see 3.4.3.2.3), twinned tone diversity (see 3.4.3.2.2), and twinned tone plus audio input diversity (see 3.4.3.2.4) operation).

1.2.3 Resident channel capability. The modulation function and the demodulation function of the VFCT terminal shall each be classified by resident channel capability. The resident channel capability for the functional mode shall be classified by channel number (see 3.4.1.1).

1.2.3.1 Modulation function resident channel capability. The modulation function resident channel capability shall be classified by the channel numbers (nos.) (see 3.4.1.1) of the modulation channels required to be operational in the VFCT terminal. This modulation function resident channel capability classification shall not preclude a VFCT terminal which is capable of providing resident channel capability for modulation Channel No. 1 through Channel No. 16 from also being classified as providing resident channel capability for other given combinations of modulation channels.

1.2.3.2 Demodulation function resident channel capability. The demodulation function resident channel capability shall be classified by the channel nos. (see 3.4.1.1) of the demodulation channels required to be operational in the VFCT terminal. This demodulation function resident channel capability classification shall not preclude a VFCT terminal which is capable of providing resident channel capability for demodulation Channel No. 1 through Channel No. 16 from also being classified as providing resident channel capability for other given combinations of demodulation channels.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery Equipment And Systems, Requirements For
MIL-C-915	Cable And Cord, Electrical, For Shipboard Use, General Specification For
MIL-C-5015	Connectors, Electrical, Circular Threaded, AN Type, General Specification For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-C-26482	Connectors, Electrical, (Circular, Miniature, Quick Disconnect, Environment Resisting), Receptacles And Plugs, General Specification For
MIL-C-28840	Connectors, Electrical, Circular Threaded, High Shock, High Density, Shipboard, Class D, General Specification For
MIL-C-28872	Cabinet, Electronic Equipment CY-4516C/S( )

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electrical And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-188-100	Common Long Haul And Tactical Communications System Technical Standard
MIL-STD-188-114	Electrical Characteristics Of Digital Interface Circuits
MIL-STD-242	Electronic Equipment Parts
MIL-STD-280	Definitions Of Item Levels, Item Exchangeability, Models, And Related Terms
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-740	Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electrical Power, Alternating Current (Metric)
DoD-STD-1679	Software Development
MIL-STD-45662	Calibration Systems Requirements

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NAVAL PERSONNEL COMMAND (NAVPERS)

NAVPERS 18068	Manual Of Navy Enlisted Manpower And Personnel Classifications And Occupational Standards

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSEA 0967-LP-239-4010	Technical Manual For Telegraph Terminal AN/UCC-1D(V)
NAVSHIPS 0967-279-4010	Technical Manual For Telegraph Terminal AN/FCC-66
NAVSHIPS 0967-292-5010	Technical Manual For Telegraph Terminal AN/FCC-67, AN/FCC-68, AN/FCC-69, AN/FCC-70, AN/FCC-71, and AN/FCC-72

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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ELECTRICAL INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77 Racks, Panels, And Associated Equipment

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The VFCT terminal shall be in accordance with MIL-E-16400 to the extent specified herein.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 VFCT terminal description. The VFCT terminal shall be comprised of the equipment, equipments, or equipment configurations necessary to conform to the requirements specified herein (see 3.4) that are applicable to the functional mode capability (see 1.2.1), in-band frequency diversity capability (see 1.2.2), and resident channel capability (see 1.2.3) classifications specified in the contract for the VFCT terminal (see 6.2).

3.4 VFCT requirements. The VFCT terminal shall conform to the requirements specified herein which are applicable to the functional mode capability, in-band frequency diversity capability, and resident channel capability classifications specified in the contract for the VFCT terminal (see 6.2).

3.4.1 General. The VFCT terminal shall conform to the requirements specified in 3.4.1.1 through 3.4.1.4.

3.4.1.1 Channel designations and associated binary frequency shift keying (FSK) audio tone frequencies. A VFCT terminal channel No. shall be assigned to each of the digital data input signals inputted to the VFCT terminal for modulation (modulation channel) and to each of the digital data output signals outputted from the VFCT terminal after demodulation (demodulation channel). The channel number designations and associated binary FSK audio tone frequencies for each channel for nondiversity operation shall be as specified in TABLE I.

TABLE I. Channel designations and associated binary FSK audio tone frequencies (Nondiversity operation, standard sense).

Channel No.	Channel mark frequency (Hz) 1/	Channel space frequency (Hz)
1	382.5	467.5
2	552.5	637.5
3	722.5	807.5
4	892.5	977.5
5	1062.5	1147.5
6	1232.5	1317.5
7	1402.5	1487.5
8	1572.5	1657.5
9	1742.5	1827.5
10	1912.5	1997.5
11	2082.5	2167.5
12	2252.5	2337.5
13	2422.5	2507.5
14	2592.5	2677.5
15	2762.5	2847.5
16	2932.5	3017.5

1/ Hertz

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3.4.1.2 Channel data signaling rate. Each modulation channel and each demodulation channel of the VFCT terminal shall be capable of operating in accordance with the requirements of this specification at a data signaling rate of 75 bits per second (bps).

3.4.1.3 Logical and signal sense. The logical and signal sense for standard sense operation of each modulation channel and each demodulation channel of the VFCT terminal shall be in accordance with the Logical and signal sense for binary signals paragraph of MIL-STD-188-100 except that the positive and negative voltage senses for mark and space shall be in accordance with MIL-STD-188-114 and except as otherwise specified herein. The VFCT terminal shall provide the capability for the operator to set the logical and signal sense for each modulation channel and each demodulation channel, in turn, for standard sense and for reverse sense operation. For reverse sense operation the logical and signal sense for the channel shall be reversed from that for standard sense operation: for example, for reverse sense operation:

a. A mark input for digital data input to the VFCT terminal modulation channel shall result in the higher binary FSK audio output tone frequency.
b. A lower binary FSK audio input tone frequency to the VFCT terminal shall result in a space digital data output for the demodulation channel.

3.4.1.4 Duty factor. The VFCT terminal shall be capable of continuous duty operation.

3.4.2 Modulator requirements. The VFCT terminal modulator shall conform to the requirements specified in 3.4.2.1 through 3.4.2.5.

3.4.2.1 Digital data input interfaces and operation. Mechanical relays and magnetic relays shall not be used in the digital data input interfaces of the modulator. Digital data input interfaces and operation of the VFCT terminal shall conform to the requirements specified in 3.4.2.1.1 through 3.4.2.1.4.

3.4.2.1.1 Data source compatibility. The modulator shall accept digital data at an isochronous data signalling rate of 75 bps from the data source equipment attached to each modulation channel digital data input and shall binary FSK modulate the data as specified in 3.4.2.1 through 3.4.2.5. Operation of the modulator with asynchronous data sources and with data sources at data signalling rates of other than 75 bps is not required.

3.4.2.1.2 Digital data input low level polar interfaces. The VFCT terminal shall provide a digital data low level polar interface conforming to the requirements of MIL-STD-188-114 for each modulation channel digital data input. Each VFCT terminal digital data low level polar interface shall be capable of, in turn, balanced and unbalanced operation.

3.4.2.1.3 Digital data input high level neutral interfaces. The VFCT terminal shall provide a digital data 60 milliampere (mA) (either side grounded) positive and, in turn, negative high level neutral interface (battery and current adjusting resistance supplied from the external loop and not by the VFCT terminal) conforming to the High level digital interface paragraph of MIL-STD-188-100 (except as otherwise specified herein) for each modulation channel digital data input. Each channel digital data input circuit shall have a resistance not exceeding 100 ohms. Each channel digital data input shall be able to withstand without damage a 120 mA continuous direct current overload (closed circuit) condition.

3.4.2.1.4 Digital data input connectors. The VFCT terminal modulator shall provide one MS3112E14-19P connector (with associated cable mating connector, MS3116E14-19S) conforming to the requirements of MIL-C-26482 for each group of eight or fewer modulation channel digital data inputs (minimum of one connector, maximum of two connectors). (For example, if eight or fewer resident channel capability is specified (see 1.2.3), a single modulator digital data input connector is required; if nine or more (16 maximum) resident channel capability is specified (see 1.2.3), two modulator digital data input connectors are required.) The modulator interfaces and pin connections to each connector shall be compatible with the KEYER LOOP INPUT connections of cabinet connector J8 as specified in NAVSEA 0967-LP-239-4010.

3.4.2.2 Modulation function. The VFCT terminal modulator shall convert multiple channel digital data input signals to frequency division multiplex (FDM) binary FSK composite VFCT channel audio output signals. The VFCT terminal modulator shall conform to the requirements specified in 3.4.2.2.1 through 3.4.2.2.3.

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3.4.2.2.1 Channel binary FSK audio output tone transmitting bias and transition distortion. With undistorted (isochronous distortion less than 0.1 percent) random mark and space digital data input to a given modulation channel at an isochronous data signalling rate of 75 bps, the time duration of the binary FSK mark tone and of the binary FSK space tone output from the VFCT terminal composite VFCT baseband audio output for the channel shall each be equal to the bit unit interval to within ±2 percent. With random mark and space digital data input to a given modulation channel with bias distortion up to and including 2 percent at an isochronous data signalling rate of 75 bps, the transmitting bias distortion of the binary FSK tones output from the VFCT terminal shall be less than or equal to the absolute magnitude of: (absolute value of digital input bias distortion +3 percent). With random mark and space digital data input to a given modulation channel with bias distortion greater than 2 percent but less than or equal to 35 percent at an isochronous data signalling rate of 75 bps, the transmitting bias distortion of the binary FSK tones outputted from the terminal shall be less than or equal to 5 percent. Operator adjustment with associated adjustment monitoring to conform to these transmitting bias distortion requirements is allowable as long as the necessary operator adjustment controls and the associated adjustment monitoring jacks are provided at the front panel of the VFCT terminal. With undistorted (isochronous distortion less than 0.1 percent) alternate mark and space digital data input to a given modulation channel at an isochronous data signalling rate of 75 bps, the average transition time distortion of the binary FSK mark and space tones output from the VFCT terminal composite VFCT baseband audio output for the channel shall be less than or equal to 1.0 percent where: the average transition time distortion is the tone on time period divided by the unit bit interval of the data signalling rate; the tone on period is (the unit bit interval of the data signalling rate) minus one-half (the mark tone to space tone transition time plus the space tone to mark tone transition time).

3.4.2.2.2 Channel binary FSK audio output frequency tone frequency tolerance and stability. The binary FSK mark and space frequency tones outputted from the VFCT terminal composite VFCT baseband audio output for a given modulation channel shall be within ±0.5 Hz of the frequency values specified in TABLE I for the channel, with stability such that deviation from the specified frequencies shall be no greater than ±1.0 Hz over a 6-month period, and with temperature stability such that total variation (minimum frequency to maximum frequency) from the binary FSK mark and space frequencies output shall be no greater than 1.0 Hz over a temperature range of from 0° Celsius (C) to 50°C.

3.4.2.2.3 Channel binary FSK audio output frequency tone level tolerance and stability. The difference in amplitude between the mark frequency tone level and the space frequency tone level for a given modulation channel binary FSK tone pair at the composite VFCT baseband audio output of the modulator after selection and initial output of the channel from the modulator (see 3.4.2.4.3) shall not exceed 2.0 decibels (dB).

3.4.2.3 Modulation diversity. The VFCT terminal modulator shall be capable of the modulation diversity operation specified in 3.4.2.3.1 and 3.4.2.3.2, except as otherwise specified herein (see 3.3).

3.4.2.3.1 Nondiversity. The VFCT terminal modulator shall provide the capability for nondiversity (untwinned modulation channel, see 1.2.2.1) operation. In the nondiversity mode of operation the modulator shall provide, at the composite VFCT baseband audio output of the VFCT terminal, modulation channel binary FSK modulated tones at the audio frequencies specified in TABLE I (standard sense operation specified in TABLE I, see 3.4.1.3 for reverse sense operation requirements) for the modulation channels that have been selected for output from the VFCT terminal modulator (see 3.4.2.4.3). Nondiversity operation of the modulation channels of the VFCT terminal modulator shall be selectable at the front panel of the VFCT terminal modulator. Channel-by-channel selection of nondiversity operation is acceptable but is not required.

3.4.2.3.2 Tone (in-band frequency) diversity. The VFCT terminal modulator shall provide the capability for tone (in-band frequency) diversity (twinned modulation channel, see 1.2.2.1) operation. In the tone (in-band frequency) diversity mode of operation the modulator shall provide, at the composite VFCT baseband audio output, modulation channel binary FSK modulated tones (two tones simultaneously outputted for each modulation channel) at the frequencies specified in TABLE II (standard sense operation specified in TABLE II - see 3.4.1.3 for reverse sense operation) for the modulation channels that have been selected for output from the VFCT terminal modulator (see 3.4.2.4.3). For a given modulation channel (and standard sense operation) a digital data input mark for the channel shall result in the modulator output of the two mark tones for the channel specified in TABLE II. Tone diversity operation of the modulation channels of the VFCT terminal modulator shall be provided by either a or b or combination thereof:

a. Operator selection at the front panel of the modulator
b. Operator selection internal to the modulator

Channel-by-channel selection of tone diversity operation is acceptable but is not required.

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TABLE II. Channel designations and associated in-band frequency diversity binary FSK audio tone frequencies (standard sense).

Channel No.	Lower frequency binary FSK tone pair		Upper frequency binary FSK tone pair	
	Channel mark frequency (Hz)	Channel space frequency (Hz)	Channel mark frequency (Hz)	Channel space frequency (Hz)
1	382.5	467.5	1742.5	1827.5
2	552.5	637.5	1912.5	1997.5
3	722.5	807.5	2082.5	2167.5
4	892.5	977.5	2252.5	2337.5
5	1062.5	1147.5	2422.5	2507.5
6	1232.5	1317.5	2592.5	2677.5
7	1402.5	1487.5	2762.5	2847.5
8	1572.5	1657.5	2932.5	3017.5

3.4.2.4 Audio output interfaces and operation. The VFCT terminal modulator audio output interfaces and operation shall be as specified in 3.4.2.4.1 through 3.4.2.4.11.

3.4.2.4.1 Composite VFCT baseband audio output designation. The modulator shall have an audio output designated audio output A which shall provide a composite baseband audio output (nominal 2800 Hz information bandwidth) of the binary FSK modulated tones for all the modulation channels tht have been selected for output from the VFCT terminal modulator (see TABLE I and TABLE II).

3.4.2.4.2 Composite VFCT baseband audio output level adjustment, tolerance, and stability. The VFCT terminal modulator shall provide the capability at the front panel of the VFCT terminal for adjustment of the composite VFCT baseband audio output over the minimum range of from -24 decibels referred to one milliwatt (dBm) to +3 dBm (average power level) when the audio output is terminated in a 600-ohm resistive load impedance. Rheostats shall not be used for adjustment of composite VFCT baseband audio output levels (or for adjustment of channel tone levels). The output level adjustment shall enable the audio output to be set, as a minimum, to each integer dBm value from -24 dBm to +3 dBm (average power level with respect to a 600-ohm resistive load impedance). Except for the immediate period of time after selection (or deselection) and output of modulation channel(s) from the modulator (see 3.4.2.4.3), the output level of the composite VFCT baseband audio output of the modulator shall be maintained to within 2 dB of the level set by the operator in the range of from -24 dBm to +3 dBm (average power level with respect to a 600-ohm resistive load impedance).

3.4.2.4.3 Modulator channel output selection (and deselection) and resulting composite VFCT baseband audio output level adjustment. The VFCT terminal modulator shall provide the capability at the front panel of the VFCT terminal for the operator to select, in turn, each modulation channel (with associated binary FSK modulated tones) of TABLE I and each modulation channel (with associated binary FSK modulated tones) of TABLE II for output at the composite VFCT baseband audio output of the VFCT terminal. The modulator shall provide the capability for the operator to deselect (remove from output at the composite VFCT baseband audio output of the VFCT terminal) each modulation channel (with associated binary FSK modulated tones) of TABLE I and each modulation channel (with associated binary FSK modulated tones) of TABLE II. The selection for output of additional modulation channels (tone(s)) and the deselection from output of existing modulation channels (tone(s)) shall be accomplished without change in the average power level of the composite VFCT baseband audio output set by the operator (except for the time period immediately following selection or deselection of a modulation channel) such that the 2 dB from set level maximum deviation specified in 3.4.2.4.2 is not exceeded (that is, the average power level of the composite VFCT baseband audio output from the modulator shall be maintained to within 2 dB of the level set by the operator for any combination of from one to 16 modulation channels (tone(s)) output from the modulator after channels are selected and deselected for output).

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3.4.2.4.4 Composite VFCT baseband audio output impedance. The modulator composite VFCT baseband audio output shall be in accordance with the Channel input/output impedances paragraph of MIL-STD-188-100, except that the frequency range shall be a minimum of 350 Hz to 3050 Hz and except as otherwise specified herein. The output shall be transformer coupled with an ungrounded center tap.

3.4.2.4.5 Differential time delay. The differential time delay through the VFCT terminal modulator of tones at each of the binary FSK audio output tone frequencies of each modulation channel (two tones for nondiversity operation and four tones for tone diversity operation) with respect to the other binary FSK audio output tone(s) of the channel shall not exceed 0.7 milliseconds (ms).

3.4.2.4.6 In-band intermodulation (IM) distortion and spurious signal levels. When 16 constant frequency (unmodulated) tones (one for each of the 16 modulation channel nondiversity binary FSK tone pairs) at, in turn, each combination of mark, space, and mark and space frequencies for the modulation channels are outputted at the modulator composite VFCT baseband audio output at composite VFCT baseband audio output levels in the range from -13 dBm to +3 dBm into a 600-ohm resistive load impedance and a single tone from a given modulation channel is removed from the output, the IM distortion products at the audio output within a ±85 Hz frequency band about the removed tone frequency shall be at least 25 dB below the level of an individual tone present. This requirement shall apply when each tone is removed, one at a time, while the other 15 tones are present. Discrete spurious signals at the modulator composite VFCT baseband audio output at frequencies less than 4125 Hz and as determined in a 1-Hz bandwidth shall be at least 25 dB below each VFCT modulation channel binary FSK tone level.

3.4.2.4.7 Harmonic distortion. The in-band and out-of-band levels of the second and higher order harmonics of each modulation channel binary FSK audio output tone at the composite VFCT baseband audio output of the modulator with the output terminated in a 600-ohm resistive load impedance shall be at least 45 dB down from the level of the tone.

3.4.2.4.8 Out-of-band spurious signal levels. Discrete spurious signals at the modulator composite VFCT baseband audio output with the output terminated in a 600-ohm resistive load impedance at frequencies greater than 4125 Hz and as determined in a 1-Hz bandwidth shall be at least 45 dB below each VFCT modulation channel binary FSK tone level.

3.4.2.4.9 Composite VFCT baseband audio output tone level tolerance and stability. The difference in amplitude between the modulation channel mark or space frequency tone having the greatest amplitude and the modulation channel mark or space frequency tone having the lowest amplitude at the modulator composite VFCT baseband audio output shall not exceed 4 dB.

3.4.2.4.10 Radio set and transmitter compatibility. The VFCT terminal modulator shall provide FDM binary FSK VFCT audio output signals to modulate radio sets and transmitters connected to the modulator composite VFCT baseband audio output.

3.4.2.4.11 Composite VFCT baseband audio output connector. The VFCT terminal modulator shall provide an MS3112E14-5P connector (with associated cable mating connector, MS3116E14-5S) conforming to the requirements of MIL-C-26482 as the composite VFCT baseband audio output connector. The modulator interfaces and pin connections to the connector shall be compatible with the PARALLEL TONES XMIT connections of cabinet connector J3 as specified in NAVSEA 0967-LP-239-4010.

3.4.2.5 Modulator interoperability with other VFCT terminals and modems. To the extent that other VFCT terminals and modems provide the demodulation, channel, and diversity capabilities necessary to interoperate with the VFCT terminal modulator as specified herein, the modulator shall operate compatibly back-to-back and shall interoperate in HF communications circuits with the demodulation functions of the VFCT terminals specified in TABLE III.

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TABLE III. Required interoperabilityof AN/UCC-1( )(V) VFCT terminal with other VFCT terminal equipments.

VFCT terminal equipment	Equipment functional mode capabilities	Equipment channel diversity capabilities	Describing document
AN/UCC-1D(V)	Modulation Demodulation	N[1/], T[2/], N, T, A[3/], T/A[4/]	5/ 5/
AN/FCC-66	Modulation Demodulation	N, T N, T, A, T/A	6/ 6/
AN/FCC-67	Modulation Demodulation	N N	7/ 7/
AN/FCC-68	Modulation Demodulation	N N	7/ 7/
AN/FCC-69	Demodulation	N	7/
AN/FCC-70	Modulation	N	7/
AN/FCC-71( )	Demodulation	N	7/
AN/FCC-72( )	Modulation	N	7/

1/ Nondiversity. All modulation functions capable of nondiversity also capable of audio output diversity with external signal splitter.
2/ Tone diversity. All modulation functions capable of tone diversity also capable of tone plus audio output diversity with external signal splitter.
3/ Audio output diversity
4/ Tone plus audio output diversity
5/ NAVSEA 0967-LP-239-4010
6/ NAVSHIPS 0967-279-4010
7/ NAVSHIPS 0967-292-5010

3.4.3 Demodulator requirements. The VFCT terminal demodulator shall conform to the requirements specified in 3.4.3.1 through 3.4.3.4.

3.4.3.1 Audio input interfaces and operation. The VFCT terminal demodulator audio input interfaces and operation shall be as specified in 3.4.3.1.1 through 3.4.3.1.11.

3.4.3.1.1 Composite VFCT baseband audio input designations. The demodulator shall have two audio inputs, designated audio input A and audio input B, for input of two separate composite VFCT baseband audio inputs (nominal 2000 Hz information bandwidth, one to 16 binary FSK modulated tones) to the demodulator for demodulation. The composite VFCT baseband audio input (A, B) or inputs (A and B) to be used for demodulation by the demodulator shall be provided by either a or b or combination thereof:

a. Operator selection at the front panel of the demodulator
b. Operator selection internal to the demodulator

Channel-by-channel selection of demodulator audio input (A, B, A and B) for demodulation is acceptable but is not required.

3.4.3.1.2 Radio set and receiver compatibility. The VFCT terminal demodulator shall accept, for demodulation in accordance with the requirements of this specification, FDM binary FSK modulated composite VFCT baseband audio output signals from radio sets and receivers connected to the demodulator composite VFCT baseband audio inputs (A, B).

3.4.3.1.3 Composite VFCT baseband audio input connectors. The VFCT terminal demodulator shall provide an MS3112E8-4P connector (with associated cable mating connector, MS3116E8-4S) conforming to the requirements of MIL-C-26482 for each of the composite VFCT baseband audio inputs (A,B). The demodulator interfaces and pin connections to the connectors shall be compatible with the PARALLEL TONES RCVR/A AND RCVR/B connections of cabinet connectors J4 and J5 as indicated in NAVSEA 0967-LP-239-4010.

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3.4.3.1.4 Internally generated spurious signal levels at composite VFCT baseband audio inputs. Discrete spurious signals generated internally to the VFCT terminal and present at one or both of the composite VFCT baseband audio inputs (A, B) of the demodulator and as determined in a 1-Hz bandwidth shall each be no greater in level than -60 dBm (with respect to a 600-ohm resistive load impedance).

3.4.3.1.5 Composite VFCT baseband audio input dynamic range. The VFCT terminal demodulator shall perform the demodulation function as specified herein with composite VFCT baseband audio input signal levels which are within the minimum range of from -25 dBm to +5 dBm (average input power level with respect to a 600-ohm resistive source impedance) and which have a maximum peak power level of +17 dBm (with respect to a 600-ohm resistive source impedance).

3.4.3.1.6 Composite VFCT baseband audio input fixed signal level. The VFCT terminal demodulator shall perform as specified herein, in turn, at fixed composite VFCT baseband audio input signal levels at its composite VFCT baseband audio inputs (A, B) of -13 dBm, -10 dBm, and 0 dBm (average power level with respect to a 600-ohm resistive source impedance).

3.4.3.1.7 Composite VFCT baseband audio input impedance. Each of the composite VFCT baseband audio inputs (A, B) of the demodulator shall be in accordance with the Channel input/output impedances paragraph of MIL-STD-188-100, except that the frequency range shall be not less than 350 Hz to 3050 Hz, except as otherwise specified herein. Each input shall be transformer coupled with an ungrounded center tap.

3.4.3.1.8 Differential time delay. The differential time delay through (added by) the demodulator for each of the binary FSK audio input tones of each demodulation channel (two tones for nondiversity operation; four tones for both tone diversity and audio input diversity operation; eight tones for tone plus audio input diversity operation) with respect to the other binary FSK audio input tones of the demodulation channel shall not exceed 0.8 ms.

3.4.3.1.9 Channel audio input transmitting bias handling capability. With random mark and space tone demodulation channel binary FSK audio inputs having transmitting bias distortion of up to and including 2 percent input to a given demodulation channel at an isochronous data signalling rate of 75 bps, the isochronous or telegraph distortion of the digital data output of the channel shall be no greater than the absolute magnitude of: (absolute value of input transmitting bias distortion +3) percent. With random mark and space tone demodulation channel binary FSK audio inputs having transmitting bias distortion of greater than 2 percent but less than or equal to 35 percent input to a given demodulation channel at an isochronous data signalling rate of 75 bps, the isochronous or telegraph distortion of the digital data output of the channel shall not exceed 5 percent. Operator adjustment with associated adjustment monitoring to conform to these transmitting bias distortion requirements is allowable as long as the necessary operator adjustment controls and the associated adjustment monitoring jacks are provided at the front panel of the VFCT terminal.

3.4.3.1.10 Channel audio input tone frequency tolerance handling capability. The demodulator shall be capable of conforming to the channel digital data output isochronous and telegraph distortion requirements of 3.4.3.3.2 with demodulation channel binary FSK audio input tones at frequencies within +3.0 Hz of the frequency values for the channel: specified in TABLE I for in turn, nondiversity (see 3.4.3.2.1) and audio input diversity (see 3.4.3.2.3) operation; specified in TABLE II for, in turn, tone diversity (see 3.4.3.2.2) and tone plus audio input diversity (see 3.4.3.2.4) operation.

3.4.3.1.11 Channel audio input tone level tolerance handling capability. The demodulator shall be capable of conforming to the channel digital data output isochronous and telegraph distortion requirements of 3.4.3.3.2 with demodulation channel binary FSK audio input mark frequency tone levels and space frequency tone levels that differ from one another by up to and including 2 dB for constant level (nonfading) conditions.

3.4.3.2 Demodulation diversity. The VFCT terminal demodulator shall convert frequency division multiplex binary FSK composite VFCT baseband audio input signals to multiple digital data output signals. The VFCT terminal demodulator shall be capable of the demodulation diversity specified in 3.4.3.2.1 through 3.4.3.2.4, except as otherwise specified herein (see 3.3).

3.4.3.2.1 Nondiversity. The VFCT terminal demodulator shall provide the capability for nondiversity (untwinned demodulation channel, see 1.2.2.2) demodulator operation. In the nondiversity mode of operation, binary FSK modulated audio tones for a given demodulation channel which correspond to the tone frequencies specified in TABLE I (standard sense operation specified in TABLE I, see 3.4.1.3 for reverse sense operation) for the demodulation channel and which are inputted to the demodulator composite VFCT baseband audio input (A, B), which has been selected by the operator to be demodulated, shall be demodulated to provide digital data output for the channel. The composite VFCT baseband audio input (A, B) to be demodulated shall be provided by either a or b or combination thereof:

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a. Operator selection at the front panel of the demodulator
b. Operator selection internal to the demodulator

Nondiversity operation shall be selectable at the front panel of the VFCT terminal demodulator. Channel-by-channel selection of nondiversity operation is acceptable but is not required.

3.4.3.2.2 Tone (in-band frequency) diversity. The VFCT terminal demodulator shall provide the capability for tone (in-band frequency) diversity (twinned demodulation channel, see 1.2.2.2) demodulator operation. In the tone diversity mode of operation, binary FSK modulated tones for a given demodulation channel which correspond to the tone frequencies specified in TABLE II (standard sense operation shown in TABLE II, see 3.4.1.3 for reverse sense operation) for the demodulation channel and which are inputted to the demodulator composite VFCT baseband audio input (A, B) which has been selected by the operator to be demodulated shall be combined and demodulated to provide digital data output for the channel. The composite VFCT baseband audio input (A, B) to be demodulated shall be provided by either a or b or combination thereof:

a. Operator selection at the front panel of the demodulator
b. Operator selection internal to the demodulator

Tone diversity operation shall be provided by either c or d or combination thereof:

c. Operator selection at the front panel of the demodulator
d. Operator selection internal to the demodulator

Channel-by-channel selection of tone diversity operation is acceptable but is not required. The diversity tone pair combination of the lower frequency and upper frequency binary FSK modulated tones shall provide a demodulated output conforming to all performance characteristics over a demodulation channel fade margin of 5.5 dB between the lower frequency and the upper frequency binary FSK modulated tones for the channel (see 3.4.3.3.8).

3.4.3.2.3 Audio input diversity. The VFCT terminal demodulator shall provide the capability for audio input diversity (untwinned demodulation channel, see 1.2.2.2) demodulator operation. In the audio input diversity mode of operation, binary FSK modulated audio tones for a given demodulation channel which correspond to the tone frequencies specified in TABLE I (standard sense operation specified in TABLE I, see 3.4.1.3 for reverse sense operation) for the demodulation channel and which are inputted to each of the demodulator composite VFCT baseband audio inputs (A and B) shall be combined and demodulated to provide digital data output for the channel. Audio input diversity operation shall be provided by either a or b or combination thereof:

a. Operator selection at the front panel of the demodulator
b. Operator selection internal to the demodulator

Channel-by-channel selection of audio input diversity operation is acceptable but is not required. The diversity pair combination of the demodulation channel binary FSK audio inputs at demodulator audio inputs A and B shall provide a demodulated output conforming to all performance characteristics over a fade margin of 5.5 dB between the demodulation channel binary FSK audio inputs at demodulator audio inputs A and B (see 3.4.3.3.9).

3.4.3.2.4 Tone plus audio input diversity. The VFCT terminal demodulator shall be capable of quadruple diversity demodulation consisting of both tone (in-band frequency) diversity (see 3.4.3.2.2) and composite VFCT baseband audio input diversity (see 3.4.3.2.3) demodulation. In the tone plus audio input diversity mode of operation, binary FSK modulated audio tones for a given demodulation channel which correspond to the tone frequencies specified in TABLE II (standard sense operation specified in TABLE II, see 3.4.1.3 for reverse sense operation) for the demodulation channel and which are inputted to each of the demodulator composite VFCT baseband audio inputs (A and B) shall be combined and demodulated to provide digital data output for the channel. Tone plus audio input diversity operation shall be provided by either a or b or combination thereof:

a. Operator selection at the front panel of the demodulator
b. Operator selection internal to the demodulator

Channel-by-channel selection of tone plus audio input diversity operation is acceptable but is not required. The tone pair diversity group (four binary FSK modulated tone pairs) shall provide a demodulated output conforming to all performance characteristics over a fade margin of 5.5 dB between binary FSK modulated tone pairs of the demodulation channel input at demodulator audio inputs A and B, and at the lower frequency and at the upper frequency binary FSK modulated tone pairs for the channel (see 3.4.3.3.10).

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3.4.3.2.5 Audio input diversity (A and B) signal time delay compensation. The demodulator shall provide the capability to compensate for unequal time delays in the separate composite VFCT baseband audio signals inputted to demodulator audio inputs A and B by either a or b:

a. Equalization of the time delay between the separate composite VFCT baseband audio signals input to demodulator audio inputs A and B
b. Compensation for or otherwise minimization of the impact upon diversity combining and bit error rate (BER) performance of the time delay between the separate composite VFCT baseband audio signals inputted to demodulator audio inputs A and B

Time delay compensation shall be provided by either c or d:

c. Automatically without operator selection or adjustment of delay equalization or compensation
d. Operator selection or adjustment of delay equalization or compensation

Channel-by-channel operator selection or adjustment of time delay equalization or compensation is acceptable but is not required. Operator selection or adjustment to provide a fixed time delay differential through the demodulator between demodulator audio inputs A and B is acceptable. The demodulator shall conform to the performance requirements specified herein for demodulation channel audio input diversity operation and for demodulation channel tone plus audio input diversity operation for each time delay in the range from zero time delay between composite VFCT baseband audio signals inputted to demodulator audio input A and to audio input B to time delays up to and including 3.5 ms between composite VFCT baseband audio signals inputted to demodulator audio input A and to audio input B.

3.4.3.3 Digital data output interfaces and operation. Mechanical relays and magnetic relays shall not be used in the demodulator digital data output interfaces. Digital data output and operation of the VFCT terminal shall conform to the requirements specified in 3.4.3.3.1 through 3.4.3.3.10.

3.4.3.3.1 Data sink compatibility. The VFCT terminal demodulator shall demodulate modulation channel tones inputted to the demodulator audio input(s) (A, B, A and B) that have been binary FSK modulated from a data source providing digital data at an isochronous data signalling rate of 75 bps and shall provide the resultant demodulation channel digital data for the input modulation channel to an isochronous data sink at a data signalling rate of of 75 bps. Operation of the demodulator with asynchronous modulation channel inputs and data sinks and with modulation channel inputs at data signalling rates of other than 75 bps is not required.

3.4.3.3.2 Channel digital data output isochronous and telegraph distortion. With undistorted (transmitting bias distortion less than 0.1 percent) random mark and space modulation channel binary FSK modulated tones inputted to the demodulator audio input(s) (A, B, A and B) at a data signalling rate of 75 bps, the time duration of the resultant mark and space digital data output from the demodulation channel for the input modulation channel shall be equal to the bit unit interval to within ± 3 percent.

3.4.3.3.3 Digital data output low level polar interfaces. The VFCT terminal shall provide a digital data low level polar interface conforming to the requirements of MIL-STD-188-114 for each demodulation channel digital data output. Each VFCT terminal digital data low level polar interface shall be capable of, in turn, balanced and unbalanced operation.

3.4.3.3.4 Digital data output high level neutral interfaces. The terminal shall provide a digital data 60 mA (either side grounded) positive and, in turn, negative high level neutral interface (battery and current adjusting resistance supplied from the external loop and not by the VFCT terminal) conforming to the High level digital interface paragraph of MIL-STD-188-100 (except as otherwise specified herein) for each demodulation channel digital data output. Each channel digital data output shall be able to withstand without damage a 120 mA continuous direct current overload (closed circuit) condition.

3.4.3.3.5 Digital data output connectors. The VFCT terminal demodulator shall provide an MS3112E14-19P connector (with associated cable mating connector, MS3116E14-19S) conforming to the requirements of MIL-C-26482 for each group of eight or fewer demodulation channel digital data outputs (minimum of one connector, maximum of two connectors). (For example, if eight or fewer resident channel capability is specified (see 1.2.3), a single demodulator digital data output connector is required; if nine or more (16 maximum) resident channel capability is specified (see 1.2.3), two demodulator digital data output connectors are required.) The demodulator interfaces and pin connections to each connector shall be compatible with the CONVERTER LOOP OUTPUT DC LOOPS output connections of cabinet connector J3 as specified in NAVSEA 0967-LP-239-4010.

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3.4.3.3.6 Demodulator BER performance for nondiversity operation with no external interference. When 16 binary FSK modulated tones (each at a data signalling rate of 75 bps) corresponding to the demodulation channels specified in TABLE I are continuously inputted to a composite VFCT baseband audio input (A, B) of the demodulator at, in turn, each fixed composite VFCT baseband audio input level in the range from -13 dBm to +3 dBm (average power level with respect to a 600-ohm resistive source impedance) for demodulation without the application of external interference (for example, no additive white Gaussian noise (AWGN), no interfering tones) and with the demodulator operating in the nondiversity mode, no greater than one bit error in the aggregate digital data output from all the 16 demodulation channel digital data outputs of the demodulator shall be experienced in a continous one-hour period.

3.4.3.3.7 Demodulator BER performance for nondiversity operation in AWGN. When 16 binary FSK modulated tones (each at a data signalling rate of 75 bps) corresponding to the demodulation channels specified in TABLE I are continuously inputted to a composite VFCT baseband audio input (A, B) of the demodulator at, in turn, each fixed composite VFCT baseband audio input level in the range from -13 dBm to 0 dBm (average power level with respect to a 600-ohm resistive source impedance) and AWGN added to the demodulator input with the given fixed composite VFCT baseband audio input level to establish, in turn, the signal to noise density ratios (SNDRs) specified in TABLE IV and with the demodulator operating in the nondiversity mode, the aggregate BER of the digital data output from all 16 demodulation channels shall not exceed the value specified in TABLE IV for the given SNDR. SNDR is defined as:

$$SNDR = \frac{P}{No} \quad Hz$$

where:

P is: the total VFCT tone audio input power for all 16 tones present
No is: the noise power per Hz of bandwidth
SNDR (in dB above 1 Hz) = 10 log (SNDR)

TABLE IV. Demodulation channel BER performance requirements for nondiversity demodulation in AWGN.

$\frac{E_b}{N_o}$ 2/ (dB)	Theoretical 16-channel SNDR (dB above 1 Hz)	Specified 16-channel SNDR for BER (dB above 1 Hz)	Channel BER shall not exceed:
8.93	39.7	42.3	1×10^{-2}
10.9	41.7	44.7	1×10^{-3}
12.3	43.1	47.1	1×10^{-4}
13.35	44.1	not specified	1×10^{-5}

1/ Energy per bit
2/ Noise power per Hz

3.4.3.3.8 Demodulator BER performance for tone diversity operation in AWGN. When 16 binary FSK modulated tones (each at a data signalling rate of 75 bps) corresponding to the eight demodulation channels specified in TABLE II are continuously inputted to a composite VFCT baseband audio input (A, B) of the demodulator at, in turn, each fixed composite VFCT baseband audio input level in the range from -13 dBm to 0 dBm (average power level with respect to a 600-ohm resistive source impedance) and AWGN is added to the demodulator input with the given fixed composite VFCT baseband audio input level to establish an SNDR and with the demodulator operating in the tone diversity mode, the demodulation channel BER performance shall be as specified herein. For each fixed composite VFCT baseband audio input level in the range of from 0 dBm to -13 dBm (average power level with respect to a 600-ohm resistive source impedance), for each demodulation channel, and for each BER in the range from 1×10^{-2} to 1×10^{-4} for the digital data output of the demodulation channel experienced for demodulation in AWGN, the SNDR required to establish the BER for tone diversity demodulation of the demodulation channel at the fixed composite VFCT baseband audio input level when the lower frequency binary FSK tone pair and

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the upper frequency binary FSK tone pair of the channel have the same signal level shall be a minimum of 1.5 dB less than the SNDR at the same fixed composite VFCT baseband audio input level required to establish the same BER for nondiversity demodulation in AWGN of, in turn, each of the two nondiversity demodulation channels associated with the lower frequency and the upper frequency binary FSK tone pairs of the tone diversity demodulation channel; that is, tone diversity demodulation shall provide a minimum 1.5 dB improvement in required SNDR over nondiversity demodulation of a demodulation channel in AWGN for BERs in the range of from 1×10^{-2} to 1×10^{-4} when the lower and upper frequency tone pair signal levels for the tone diversity channel are equal in level. For each fixed composite VFCT baseband audio input level in the range of from 0 dBm to -13 dBm (average power level with respect to a 600-ohm resistive source impedance), for each demodulation channel, and for each BER in the range from 1×10^{-2} to 1×10^{-4} for the digital data output of the demodulation channel experienced for demodulation in AWGN, the SNDR required to establish the BER for tone diversity demodulation of the demodulation channel at the fixed composite VFCT baseband audio input level when the tone pair signal levels of one of the binary FSK tone pairs for the channel (lower frequency, upper frequency) is reduced by 5.5 dB with respect to the tone pair signal levels of the other tone pair shall be no greater than the SNDR at the same fixed composite VFCT baseband audio input level required to establish the same BER for nondiversity demodulation in AWGN of, in turn, each of the two nondiversity demodulation channels associated with the lower frequency and the upper frequency binary FSK tone pairs of the tone diversity demodulation channel; that is, tone diversity demodulation shall provide equal performance (as a minimum) in required SNDR over nondiversity demodulation of a demodulation channel in AWGN for BERs in the range of from 1×10^{-2} to 1×10^{-4} when, in turn, each of the binary FSK tone pair signal levels (lower frequency, upper frequency) for the channel is reduced up to and including 5.5 dB in level with respect to the other.

3.4.3.3.9 Demodulator BER performance for audio input diversity operation in AWGN. When 16 binary FSK modulated tones (each at a data signalling rate of 75 bps) corresponding to the 16 demodulation channels specified in TABLE I are continuously inputted to both the composite VFCT baseband audio inputs (A and B) of the demodulator at, in turn, each fixed composite VFCT baseband audio input level in the range of from -13 dBm to 0 dBm (average power level with respect to a 600-ohm resistive source impedance) and AWGN is added to the demodulator inputs with the given fixed composite VFCT baseband audio input level to establish an SNDR and with the demodulator operating in the audio input diversity mode, the demodulation channel BER performance shall be as specified herein. For each fixed composite VFCT baseband audio input level in the range of from 0 dBm to -13 dBm (average power level with respect to a 600-ohm resistive source impedance), for equal composite VFCT baseband audio input signal levels at each demodulator audio input (A and B), for each demodulation channel, and for each BER in the range from 1×10^{-2} to 1×10^{-4} for the digital data output of the demodulation channel experienced for demodulation in AWGN, the SNDR required to establish the BER for audio input diversity demodulation of the demodulation channel at the fixed composite VFCT baseband audio input levels when the signal levels of the channel binary FSK tone pair at demodulator audio input A are equal to the signal levels of the channel binary FSK tone pair at demodulator audio input B shall be a minimum of 1.5 dB less than the SNDR at the same fixed composite VFCT baseband audio input level required, in turn, at each demodulator audio input (A, B) to establish the same BER for nondiversity demodulation of the channel in AWGN; that is, audio input diversity demodulation shall provide a minimum 1.5 dB improvement in required SNDR over nondiversity demodulation of a demodulation channel in AWGN for BERs in the range of from 1×10^{-2} to 1×10^{-4} when the signal levels for the binary FSK tone pairs at demodulator audio inputs A and B for the channel are equal. For each fixed composite VFCT baseband audio input signal in the range of from 0 dBm to -13 dBm (average power level with respect to a 600-ohm resistive source impedance), for fixed composite VFCT baseband audio input signals that are the equal in level at each demodulator audio input (A and B), for each demodulation channel, and for each BER in the range of from 1×10^{-2} to 1×10^{-4} for the digital data output of the demodulation channel experienced for demodulation in AWGN, the SNDR at each demodulator audio input (A and B) required to establish the BER for audio input diversity demodulation of the demodulation channel when the signal levels of the channel binary FSK tone pair at, in turn, each of the audio inputs of the demodulator is 5.5 dB less than the signal levels of the channel binary FSK tone pair at the other demodulator audio input shall be no greater than the SNDR at the same fixed composite VFCT baseband audio input level at, in turn, each demodulator audio input (A, B) required to establish the BER for nondiversity demodulation of the channel in AWGN; that is, audio input diversity demodulation shall provide equal performance (as a minimum) in required SNDR over nondiversity demodulation of a given channel in AWGN for BERs in the range of from 1×10^{-2} to 1×10^{-4} when the signal levels of, in turn, each of the binary FSK tone pairs of the channel at each of the demodulator audio inputs (A, B) is reduced up to and including 5.5 dB in level with respect to the signal levels of the other binary FSK tone pair of the channel at the other demodulator audio input.

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3.4.3.3.10 Demodulator BER performance for tone plus audio input diversity operation in AWGN. When 16 binary FSK modulated tones (each at a data signalling rate of 75 bps) corresponding to the eight demodulation channels specified in TABLE II are continuously inputted to both the composite VFCT baseband audio inputs (A and B) of the demodulator at, in turn, each fixed composite VFCT baseband audio input level in the range of from -13 dBm to 0 dBm (average power level with respect to a 600-ohm resistive source impedance) and AWGN is added to the demodulator audio inputs with the given fixed composite VFCT baseband audio input level to establish an SNDR and with the demodulator operating in the tone plus audio input diversity mode, the demodulation channel BER performance shall be as specified herein. For each fixed composite VFCT baseband audio input level in the range of from 0 dBm to -13 dBm (average power level with respect to a 600-ohm resistive source impedance), for equal composite VFCT baseband audio input signal levels at each demodulator audio input (A and B), for each demodulation channel, and for each BER in the range of from 1×10^{-2} to 1×10^{-4} for the digital data output of the demodulation channel experienced for demodulation in AWGN, the SNDR required to establish the BER for tone plus audio input diversity demodulation of the demodulation channel at the fixed composite VFCT baseband audio input levels when the binary FSK tone pair signal levels for each of the four binary FK tone pairs of the channel are equal in level shall be a minimum of 3.0 dB less than the SNDR at the same fixed composite VFCT baseband audio input level required to establish the same BER for nondiversity demodulation, in turn, of each of the four binary FSK tone pairs of the channel in AWGN; that is, tone plus audio input diversity demodulation shall provide a minimum 3.0 dB improvement in required SNDR over nondiversity demodulation of a demodulation channel in AWGN for BERs in the range of from 1×10^{-2} to 1×10^{-4} when the signal levels of each of the four binary FSK tone pairs of the channel are equal in level. For each fixed composite VFCT baseband audio input level in the range of from 0 dBm to -13 dBm (average power level with respect to a 600-ohm resistive source impedance), for fixed composite VFCT baseband audio input signals of equal level at each demodulator audio input (A and B), for each demodulation channel, and for each BER in the range of from 1×10^{-2} to 1×10^{-4} for the digital data output of the demodulation channel experienced for demodulation in AWGN, the SNDR at each demodulator audio input (A and B) required to establish the BER for tone plus audio input diversity demodulation of the demodulation channel when the signal levels of three of the four binary FSK tone pairs of the channel are each 5.5 dB lower then the remaining binary FSK tone pair of the channel shall be no greater than the SNDR at the same fixed composite VFCT baseband audio input level at, in turn, each demodulator audio input (A, B) required to establish the same BER for nondiversity demodulation of the demodulation channel in AWGN; that is, tone plus audio input diversity demodulation shall provide equal performance (as a minimum) in required SNDR over nondiversity demodulation of a demodulation channel in AWGN for BERs in the range of from 1×10^{-2} to 1×10^{-4} when the signal levels of three of the four binary FSK tone pairs of the tone plus audio diversity demodulation channel are each reduced up to and including 5.5 dB in level with respect to the signal level of the remaining demodulation channel binary FSK tone pair.

3.4.3.4 Demodulator interoperability with other VFCT terminals and modems. To the extent that other VFCT terminals and modems provide the modulation, channel, and diversity capabilities necessary to interoperate with the VFCT terminal demodulator as specified herein, the demodulator shall operate compatibly back-to-back and shall interoperate in HF communications circuits with the modulation functions of the VFCT terminals specified in TABLE III.

3.4.4 Integrated modulator and demodulator requirements. The VFCT terminal with modulation and demodulation functional mode capability (see 1.2.1) shall conform to the additional requirements of 3.4.4.1 through 3.4.4.5. The VFCT terminal with modulation and demodulation functional mode capability shall have the capability for connecting the VFCT terminal modulator and the VFCT terminal demodulator back-to-back (for example, by means of external cabling).

3.4.4.1 Back-to-back nondiversity operation BER performance with no external interference. When the composite VFCT baseband audio output of the VFCT terminal modulator is connected back-to-back, in turn, with each of the composite VFCT baseband audio inputs (A, B) of the VFCT terminal demodulator and the combined modulator and demodulator operated without external interfering signals as specified in 3.4.3.3.6, the BER performance of the back-to-back configuration shall conform to the requirements of 3.4.3.3.6.

3.4.4.2 Back-to-back nondiversity operation BER performance in AWGN. When the composite VFCT baseband audio output of the VFCT terminal modulator is connected back-to-back, in turn, with each of the composite VFCT baseband audio inputs (A, B) of the VFCT terminal demodulator and the combined modulator and demodulator operated with AWGN applied with the modulator output signal to the demodulator audio input as specified in 3.4.3.3.7, the BER performance of the back-to-back combination shall conform to the requirements of 3.4.3.3.7.

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3.4.4.3 Back-to-back tone-diversity operation BER performance in AWGN. When the composite VFCT baseband audio output of the VFCT terminal modulator is connected back-to-back, in turn, with each of the composite VFCT baseband audio inputs (A, B) of the VFCT terminal demodulator and the combined modulator and demodulator operated with AWGN applied with modulator output signal to the demodulator audio input as specified in 3.4.3.3.8, the BER performance of the back-to-back combination shall conform to the requirements of 3.4.3.3.8.

3.4.4.4 Back-to-back channel digital data output isochronous and telegraph distortion. When the composite VFCT baseband audio output of the VFCT terminal modulator is connected back-to-back, in turn, with each of the composite VFCT baseband audio inputs (A, B) of the VFCT terminal demodulator, a composite VFCT baseband audio input level of, in turn, +3 dBm and -13 dBm (average power level with respect to a 600-ohm resistive source impedance) is applied to the demodulator input, and the combined modulator and demodulator operated in the nondiversity demodulation mode as specified in TABLE I with 16 undistorted (isochronous and telegraph distortion less than 0.1 percent) random mark and space digital data inputs applied to the 16 modulation channels of the modulator, the maximum isochronous distortion of the digital data from each of the 16 demodulation channel digital data outputs shall not exceed +4 percent and the maximum total peak telegraph distortion from each demodulation channel digital data output shall not exceed 5 percent.

3.4.4.5 Back-to-back channel crosstalk limits. When the composite VFCT baseband audio output of the VFCT terminal modulator is connected back-to-back, in turn, with each of the composite VFCT baseband audio inputs (A, B) of the VFCT terminal demodulator, a composite VFCT baseband audio input level of 0 dBm (average power level with respect to a 600-ohm resistive load impedance) is applied to the demodulator audio input, and the combined modulator and demodulator operated in the nondiversity mode as specified in TABLE I with 16 undistorted (isochronous and telegraph distortion less than 0.1 percent) random mark and space digital data inputs applied to the 16 modulation channels of the modulator but with the binary FSK modulated tones of a given modulation channel inputted to the audio input of the demodulator attenuated by 25 dB with respect to the tone levels of the other 15 modulation channels inputted to the audio input of the demodulator, the telegraph disortion of the digital data output of the attenuated modulation channel from the associated demodulation channel digital data output of the demodulator shall not exceed 40 percent total peak distortion. Conformance to this requirement shall be achieved when the attenuated modulation channel is, in turn, each of the 16 modulation channels inputted to the demodulator.

3.5 Physical characteristics. Physical characteristics of the VFCT terminal shall be as specified in 3.5.1 and 3.5.2.

3.5.1 Weight. The maximum allowable weights for 16-channel VFCT terminals shall not exceed the weights specified in TABLE V.

TABLE V. Maximum allowable VFCT terminal weight.

VFCT terminal	Capability	Weight (lbs) 1/
A	Modulation and demodulation functional mode capability; untwinned and twinned channel capability for both modulation and demodulation functions; Channel No. 1 to Channel No. 16 (untwinned) and Channel No. 1 to Channel No. 8 (twinned) resident channel capability for both modulation and demodulation functions	300
B	Demodulation functional mode capability; untwinned and twinned channel capability; Channel No. 1 to Channel No. 16 (untwinned) and Channel No. 1 to Channel No. 8 (twinned) resident channel capability	185

1/ Pounds

3.5.2 Dimensions. The maximum allowable dimensions for 16-channel VFCT terminals shall not exceed the dimensions specified in TABLE VI.

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TABLE VI. Maximum allowable VFCT terminal dimensions.

VFCT terminal	Height (in.) 1/	Case width (in.)	Rack and cabinet installation width (in.)	Total depth (in.)	Rack and cabinet installation depth (behind front panel) (in.)
A 2/	28.0	17.375	19.0	23.5	19.5
B 3/	14.0	17.375	19.0	23.5	19.5

1/ Inches
2/ Modulation and demodulation functional mode capability; untwinned and twinned channel capability for both modulation and demodulation functions; Channel No. 1 to Channel No. 16 (untwinned) and Channel No. 1 to Channel No. 8 (twinned) resident channel capability for both modulation and demodulation functions
3/ Demodulation functional mode capability; untwinned and twinned channel capability; Channel No. 1 to Channel No. 16 (untwinned) and Channel No. 1 to Channel No. 8 (twinned) resident channel capability

3.6 Human engineering. Human engineering shall be as specified in the Human engineering paragraph of MIL-E-16400.

3.6.1 Operational personnel. The maximum operator skill levels required for operation of the VFCT terminal shall not exceed those required of a Radioman with an E-4 (PO3) rating and training similar to that needed to acquire the Navy Enlisted Classification (NEC) 2318 (as specified in NAVPERS 18063) or those required of an Operations Specialist with an E-4 (PO3) rating and training similar to that needed to acquire the NEC 0312 (as specified in NAVPERS 18068).

3.7 Environmental service conditions. The VFCT terminal shall maintain specified operational performance under the environmental service conditions specified in 3.7.1 through 3.7.6. The VFCT terminal shall operate within tolerances throughout the specified conditions without alignment or adjustment other than accessible controls normally employed for operation of the VFCT terminal.

3.7.1 Temperature. The VFCT terminal shall maintain specified operational performance when subjected to temperatures throughout the complete operating temperature Range 4 of MIL-E-16400. The VFCT terminal shall not be damaged nor shall operational performance be degraded when restored to the operating temperature Range 4 of MIL-E-16400 after being subjected to the complete nonoperating temperature Range 4 of MIL-E-16400.

3.7.2 Humidity. The VFCT terminal shall conform to the requirements of the Humidity paragraph of MIL-E-16400.

3.7.3 Salt fog. The VFCT terminal shall conform to the requirements of the Salt fog (spray) paragraph of MIL-E-16400 for sheltered equipment.

3.7.4 Shock. The VFCT terminal shall be capable of withstanding the Grade A, Class I, lightweight, Type A shock test of MIL-S-901 without interruption or degradation of service beyond the duration of the shock impact when mounted as specified in 3.12.1 and 4.6.5.5.

3.7.5 Vibration. The VFCT terminal shall withstand without damage and shall maintain specified operational performance when being subjected to the Type I vibration test of MIL-STD-167-1 when mounted as specified in 3.12.1 and 4.6.5.6.

3.7.6 Inclination. The VFCT terminal shall withstand without damage and shall maintain specified operational performance when subjected to the 45-degree inclination angle test specified in MIL-E-16400 when mounted as specified in 3.12.1 and 4.6.5.7.

3.8 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials, and processes paragraph of MIL-E-16400 except as specified in 3.8.1 through 3.8.1.7.

3.8.1 Parts. Parts shall be as specified in 3.8.1.1 through 3.8.1.7.

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3.8.1.1 Nonstandard parts approval. Nonstandard parts approval shall be in accordance with the Nonstandard parts paragraph of MIL-E-16400 and as specified herein. All nonstandard parts shall be approved by the procuring activity prior to final inspection and tests. Nonstandard parts submitted for approval should be available from two or more independent sources which shall be identified.

3.8.1.2 Interchangeability. Interchangeability shall be in accordance with the Interchangeability paragraph of MIL-E-16400 and as specified herein. Like units, assemblies, subassemblies, modules, printed circuit cards, and replaceable parts for a given VFCT terminal configuration with given functional mode capability (see 1.2.1), in-band frequency diversity capability (see 1.2.2), and resident channel capability (see 1.2.3), shall be physically, electrically, and functionally interchangeable with the corresponding units of the same configuration produced in accordance with the requirements of this specification. Definitions of assemblies, subassemblies, and so forth shall be as specified in MIL-STD-280.

3.8.1.3 Batteries. Batteries shall not be used in the VFCT terminal without the approval of the procuring activity.

3.8.1.4 External interface connectors. External (nonradio frequency (RF)) interface connectors (and associated external cable compatibility) not specified elsewhere in this specification shall be selected in accordance with Part 8 of MIL-STD-242, except as otherwise approved by the procuring activity. (Part 8 of MIL-STD-242 gives first priority to MIL-C-28840 connectors with second priority to MIL-C-5015 connectors.)

3.8.1.5 Microelectronic devices. Unless otherwise approved by the procuring activity, microelectronic devices shall be in accordance with the requirements of the Microelectronic devices paragraph of MIL-E-16400 and as specified herein. Unless otherwise approved by the procuring activity, all microelectronic devices used in the VFCT terminal shall be available from more than one commercial manufacturer.

3.8.1.5.1 Custom microelectronic devices. Custom microelectronic devices (for example, read only memory (ROM)) shall not be used in the VFCT terminal without the approval of the procuring activity.

3.8.1.6 Semiconductor devices. Unless otherwise approved by the procuring activity, semiconductor devices shall be in accordance with the requirements of the Semiconductor devices paragraph of MIL-E-16400 and as specified herein. Germanium semiconductor devices shall not be used in the VFCT terminal. Unless otherwise approved by the procuring activity, all semiconductor devices used in the VFCT terminal shall be available from more than one commercial manufacturer.

3.8.1.7 Electron tubes. Electron tubes shall not be used in the VFCT terminal.

3.9 Electrical design and construction. Electrical design and construction shall be as specified in 3.9.1 through 3.9.1.5.

3.9.1 Alternating current (AC) source power. AC source power shall be as specified in 3.9.1.1 through 3.9.1.4.

3.9.1.1 AC source power interface. The VFCT terminal shall be designed to operate from single-phase 115 volt alternating current (VAC) root-mean-square (rms) (±10 percent) AC source power, in turn, of 50 Hz (±5 percent) and 60 Hz (±5 percent) line frequency and shall conform to the Steady-state voltage and frequency, Transient voltage, Transient frequency, and Power and power factor paragraphs of MIL-E-16400. The AC source power connector(s) for the VFCT terminal shall be either an MS3112E12-3P connector (with associated cable mating connector, MS3116E12-3S) conforming to the requirements of MIL-C-26482 or a plug conforming to the Class DJ, Series 3400 requirements of MIL-C-5015 (MS3406DJ15E2P/S) with associated Class D series receptacle (with backshell) for attachment to a 3SJ16 cable as specified in MIL-C-915. Assignment of VFCT terminal AC source power interface pins, conductors, and conductor insulation colors within the unit shall be as specified in TABLE VII.

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TABLE VII. VFCT terminal AC source power connection pin assignments and conductor color codes.

AC source power pin designation	Conductor assignment	Conductor insulation color
A	115 VAC return	White or light blue
B	Safety ground	Green or green with yellow stripe
C	115 VAC high (supply)	Black or brown

3.9.1.2 AC source input power. The maximum allowable AC source input power for 16-channel VFCT terminals shall not exceed the values specified in TABLE VIII.

TABLE VIII. Maximum allowable VFCT terminal AC source input power.

VFCT terminal	Capability	Power (watts)
A	Modulation and demodulation functional mode capability; untwinned and twinned channel capability for both modulation and demodulation functions; Channel No. 1 to Channel No. 16 (untwinned) and Channel No. 1 to Channel No. 8 (twinned) resident channel capability for both modulation and demodulation functions	230
B	Demodulation functional mode capability; untwinned and twinned channel capability; Channel No. 1 to Channel No. 16 (untwinned) and Channel No. 1 to Channel No. 8 (twinned) resident channel capability	230

3.9.1.3 AC source power ON/OFF switch and indicator. A switch for disconnecting the equipment from all AC source electrical power systems shall be mounted on the VFCT terminal equipment front panel and its function clearly labeled. The switch shall break all power conductors (except the safety ground conductor) of the power circuit when in the OFF position. When the switch is in the ON position, the equipment shall be fully energized. A green indicator mounted near or as part of the AC source power ON/OFF switch shall be lighted when the equipment is energized from AC source power.

3.9.1.4 Electrical overload protection. Electrical overload protection shall be in accordance with the requirements of the Electrical overload protection paragraph of MIL-E-16400 and as specified herein. If fuses are used for electrical overload protection, blown fuse indicators visible on the front panel of the equipment shall be provided.

3.9.1.5 Resumption of operation after AC source power interruption. The VFCT terminal shall return to operation in accordance with the performance requirements specified herein without operator intervention within 1 second after the conclusion of AC source power interruptions at the equipment AC source power input connector of durations specified in the Power interruption paragraph of MIL-E-16400 (and of durations in the range of from 0.5 seconds to 120 seconds). Return to operation shall be in the same operating condition and status existing prior to the AC source power interruption.

3.10 Electromagnetic compatibility (EMC). The VFCT terminal shall be self-compatible and shall conform to the requirements of MIL-STD-461 for Class A4 modem equipment to the extent specified herein. The VFCT terminal shall conform to the RE02 radiated emissions requirements of MIL-STD-461 for Class A4 modem equipment except that the RE02 limits and frequency ranges for narrowband and for broadband emission limits shall be as specified in FIGURE 1 and FIGURE 2, respectively. The AC leads of the VFCT terminal shall conform to the CE03 conducted emissions requirements of MIL-STD-461 for Class A4 modem equipment except that the CE03 limits and frequency ranges for narrowband and for broadband emission limits shall be as specified in FIGURE 3 and FIGURE 4, respectively.

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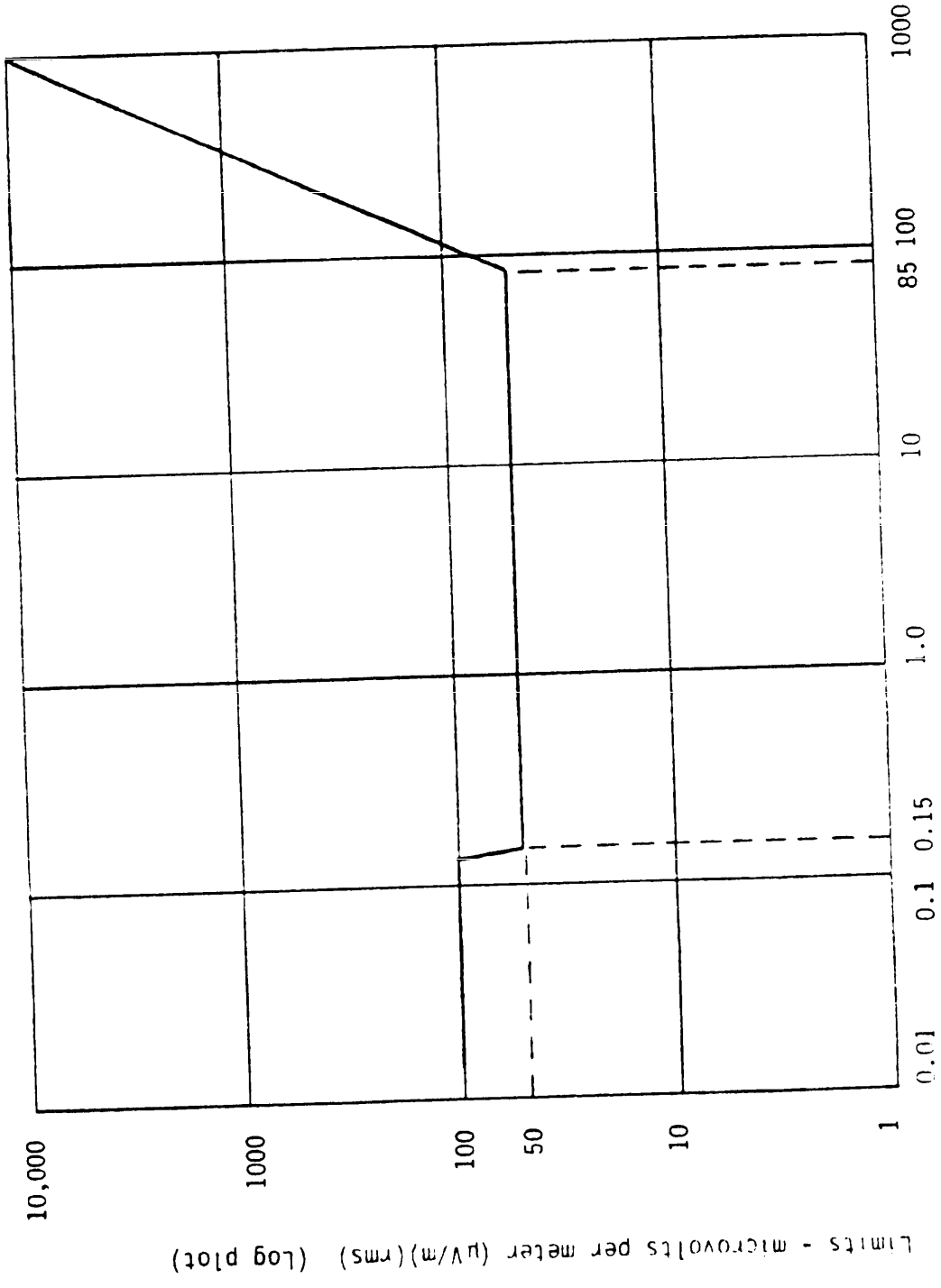


FIGURE 1. Limit for RE02 narrowband emissions.

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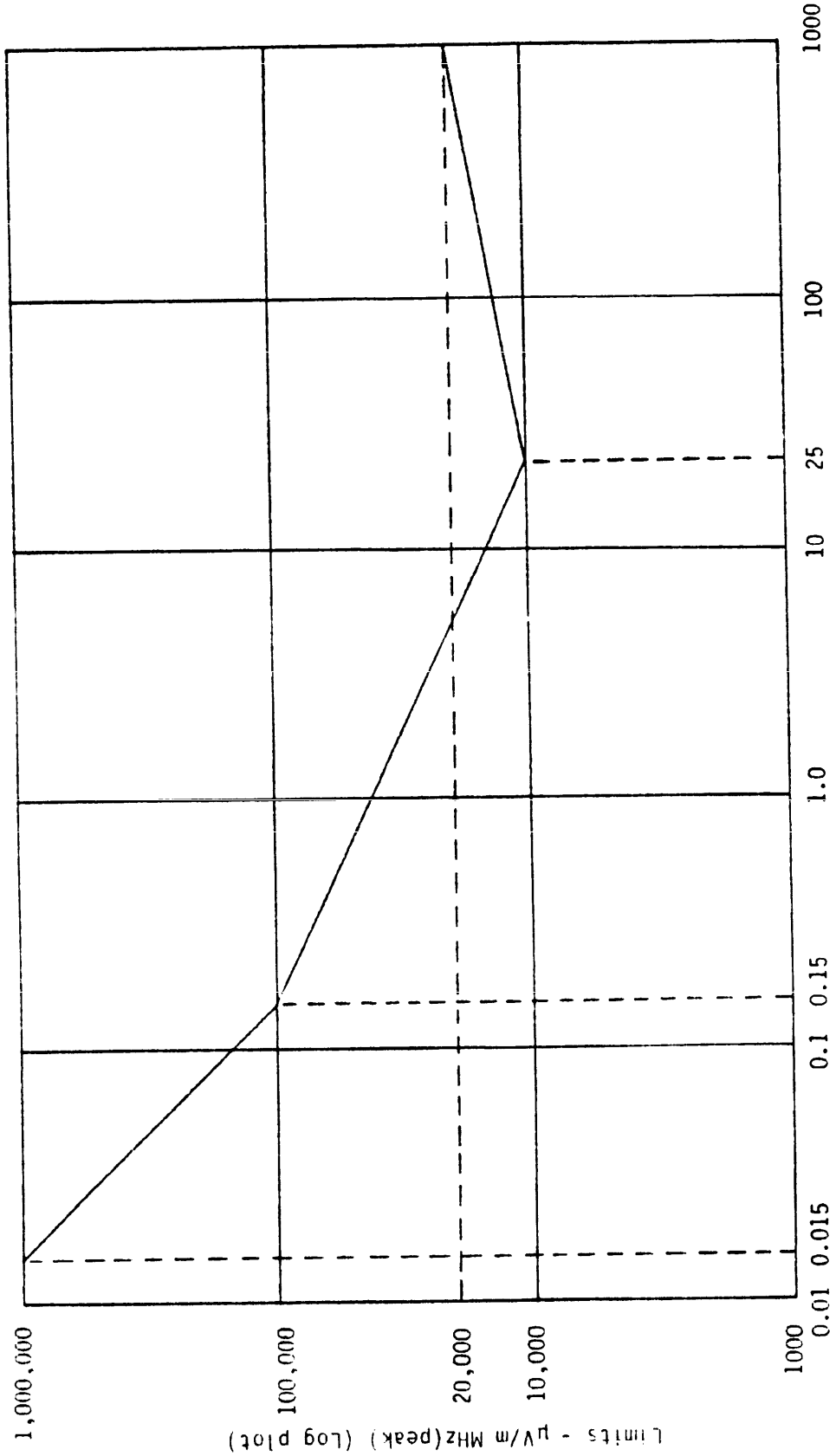


FIGURE 2. Limit for RE02 broadband emissions.

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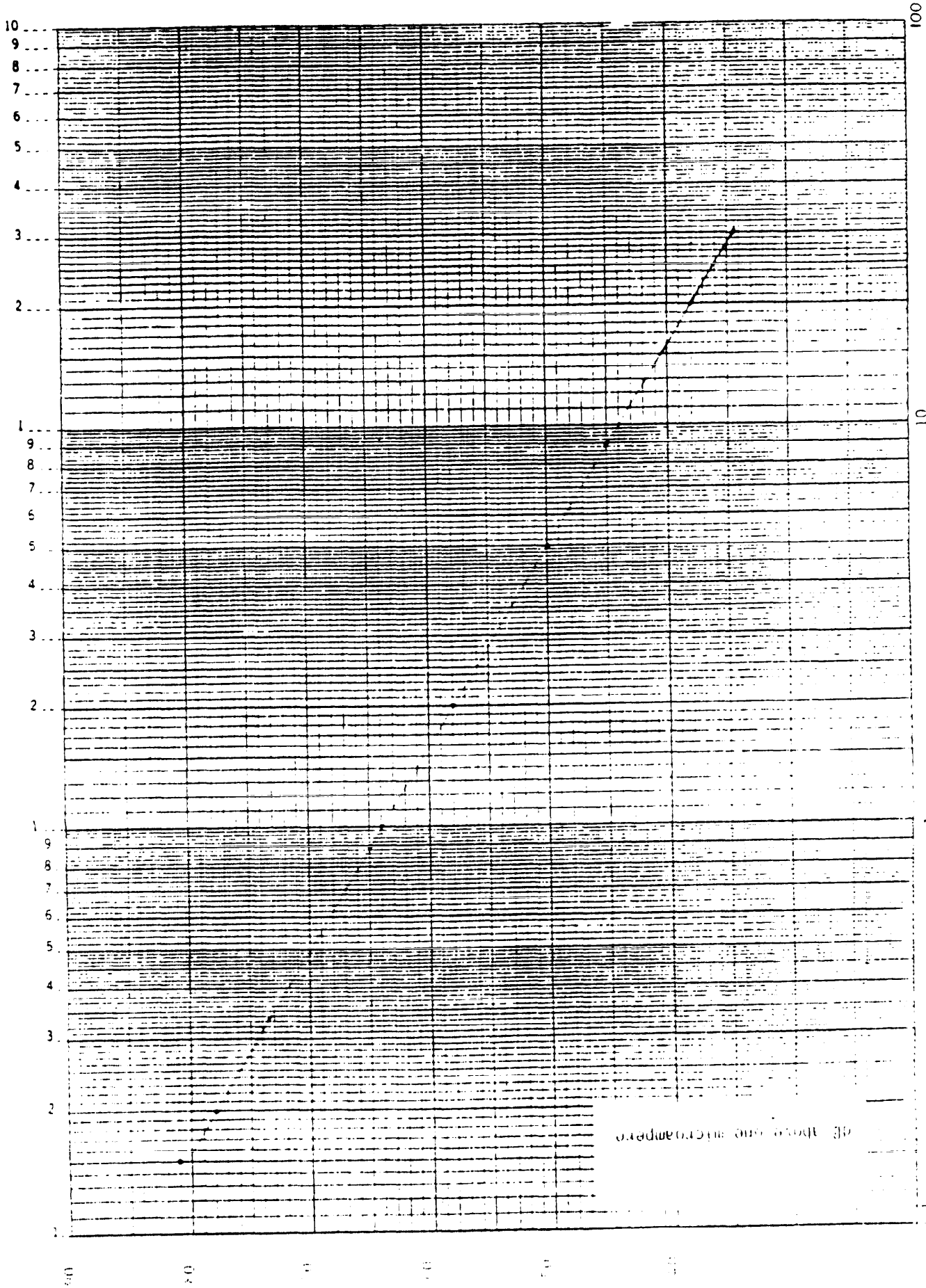


FIGURE 3. Limit for CE03 narrowband emissions, AC leads.

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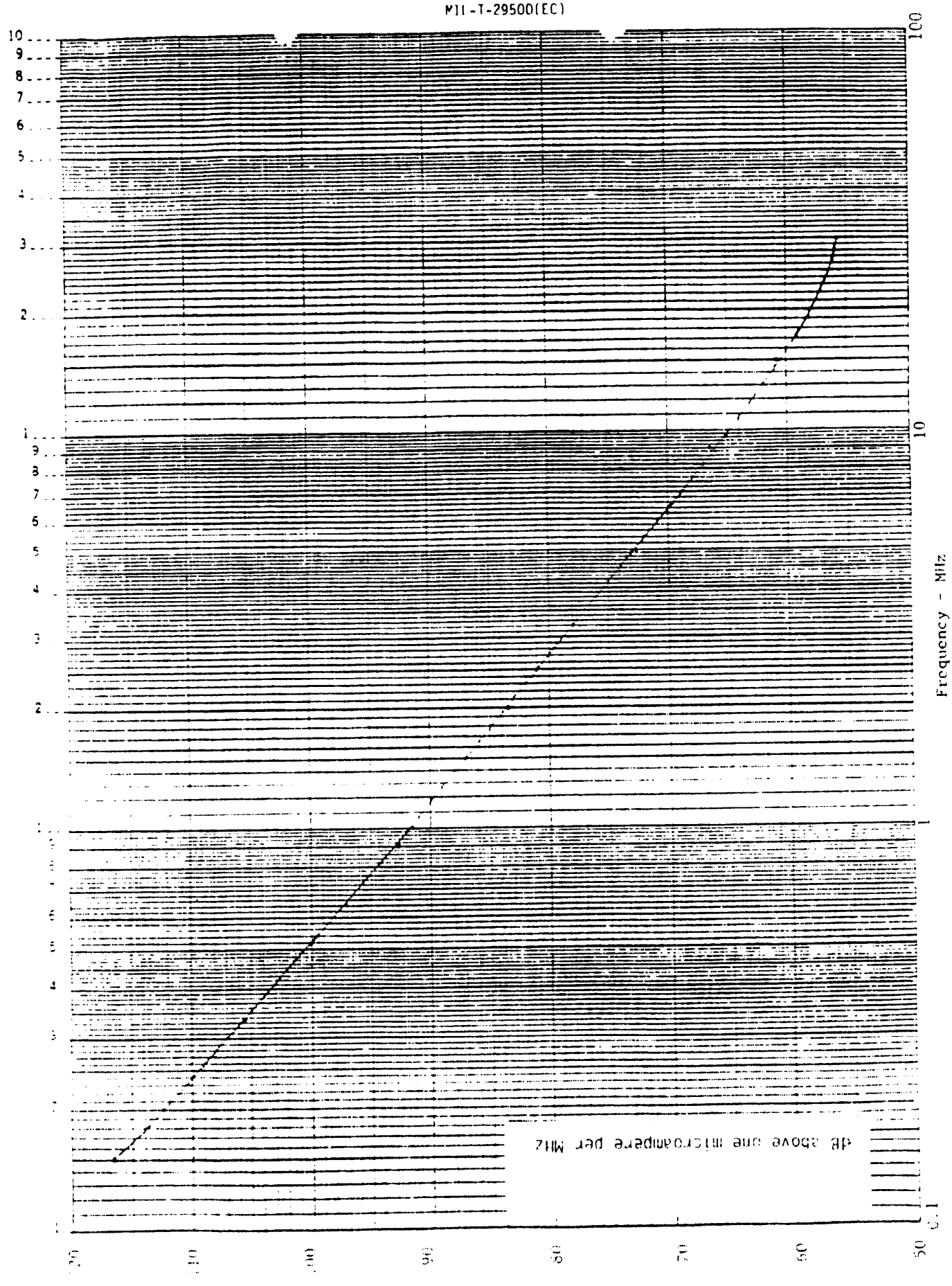


FIGURE 4. Limit for CE03 broadband emissions, AC leads.

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3.11 Maintenance design. Maintenance design shall be as specified in 3.11.1 through 3.11.7.

3.11.1 Maintenance concept. The maintenance concept for the VFCT terminal is for most maintenance to be performed at the organizational level of maintenance. The equipments shall be modularly designed and constructed to maximize the ability of organizational level maintenance personnel to restore the equipment to full operational capability by fault isolation to and replacement of the line replaceable units (LRUs) from authorized spares. Preventive maintenance will also be performed at the organizational level of maintenance.

3.11.2 Organizational level maintenance personnel. The maximum skill level required for routine preventive and corrective maintenance of the VFCT terminal at the organizational level of maintenance shall not exceed those required of an Electronics Technician with an E-4 (PO3) rating and training similar to that needed to acquire the NEC 1453 (as specified in NAVPERS 18068) or those required of a Data Processing Technician with an E-4 (PO3) rating and training similar to that needed to acquire the NEC 2707 (as specified in NAVPERS 18068).

3.11.3 General purpose electronic test equipment (GPETE). GPETE, to be used at all levels of maintenance for the VFCT terminal, shall be selected from the standard test equipments specified in MIL-STD-1364.

3.11.4 Special tools. The VFCT terminal shall be designed to preclude the need for special tools at the organizational level of maintenance. The design of the equipment shall be such that, as a maximum, a screwdriver is the only tool required for installation, calibration, alignment, configuration, and any combination thereof (see 6.4).

3.11.5 Self-test or built-in-test (BIT). The VFCT terminal shall have the capability to detect at least 80 percent of all failures or malfunctions using self test or BIT techniques, or both. Isolation of the failures should be performed using meters, lights, and other indicators which are an integral part of the VFCT terminal, or by using external GPETE, special purpose electronic test equipment (SPETE), or both. SPETE, if used, shall consist of a single equipment which shall be supplied with each VFCT terminal. (The use of SPETE to conform to this requirement is not preferred and shall be subject to procuring activity approval.) SPETE shall not be included in the weight, dimensions, and AC source power requirements of 3.5.1, 3.5.2, and 3.9.1.2.

3.11.6 Protection from BIT and built-in-test equipment (BITE) failures. The ability for the VFCT terminal to perform its intended function shall not be impaired by the malfunction of BIT or BITE.

3.11.7 Test point provisions. Test point provisions shall be as specified in a through c:

a. Signal quality and circuit performance. Test points, test jacks, or both, should be provided to permit the injection of signals and the monitoring of signals at the input/output (I/O) terminals of the assemblies, subassemblies, modules, and of printed circuit cards. Test points critical to fault isolation to the LRU using GPETE or SPETE should be accessible without the use of extender boards or cables.

b. Fault isolation. Test points, test jacks, or both, shall be provided to permit the injection and measurement of signals into the assemblies, subassemblies, modules, and printed circuit cards for isolating all faults to the electrical functions (for example, integrated circuit, filter, mixer, voltage controlled oscillator, and so forth). Test points and test jacks should provide the means for performing fault isolation on the printed circuit card with the card removed from the equipment.

c. Alignment. Test points, test jacks, or both, should be provided to permit alignment of the equipment assemblies and subassemblies. The requirement for alignment of the VFCT terminal should be minimized.

3.12 Mechanical design and construction. Mechanical design and construction shall be as specified in in 3.12.1 through 3.12.3.

3.12.1 Installation. The VFCT terminal shall be designed to be capable of being rack-panel mounted in a CY-4516( )/S cabinet as specified in MIL-C-28872 and of being rack-panel mounted in an unenclosed rack. The VFCT terminal shall be provided with mounting hardware which will enable the equipment to be mounted at the front of the cabinet or rack as a minimum. The front mounting of the VFCT terminal with mounting hardware shall conform to the requirements of EIA RS-310-C-77. The height(s) of the VFCT terminal units or units installed with associated cabinet and rack mounting hardware shall conform to the standard height requirements specified in EIA RS-310-C-77. All hardware necessary for installation of the VFCT terminal in a CY-4516( )/S cabinet shall be supplied with and considered a part of the VFCT terminal.

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3.12.2 Enclosures. VFCT terminal enclosures shall conform to MIL-STD-454, Requirement 55. As a minimum the degree of enclosure shall be dripproof (15 degrees) protected in accordance with MIL-STD-108.

3.12.3 Airborne and structureborne noise. The VFCT terminal shall conform to airborne noise acceptance limits 10 dB less (10 dB or more stringent) than those limits specified in MIL-STD-740 for Grade A, Type 3 equipment. The VFCT terminal shall conform to the structureborne noise acceptance limits specified in MIL-STD-740 for Grade A, Type 3 equipment.

3.13 Thermal design. Thermal design shall be in accordance with 3.13.1 through 3.13.5.

3.13.1 Forced air cooling. Forced air cooling, if used, shall be as specified in the Forced air-cooling paragraph of MIL-E-16400 and shall be compatible with the installation of the VFCT terminal in a CY-4516( )/S cabinet (which is forced air cooled); the VFCT terminal shall comply with the thermal design requirements specified herein when installed in a CY-4516( )/S cabinet. Modification of the CY-4516( )/S cabinet (for example, cut out of the rear cabinet panel) will be acceptable as an alternative for installation of a forced air cooled VFCT terminal in a forced air cooled CY-4516( )/S cabinet.

3.13.2 Water cooling. Water cooling of the VFCT terminal or any of its components shall not be used.

3.13.3 Ventilation openings. Ventilation openings, if required for cooling, shall be as specified in the Ventilation openings paragraph of MIL-E-16400.

3.13.4 Air filters. Air filters, if supplied, shall be as specified in the Air filters paragraph of MIL-E-16400.

3.13.5 Cooling during maintenance. Cooling during maintenance shall be as specified in the Cooling during maintenance paragraph of MIL-E-16400.

3.14 Safety. Safety shall be in accordance with the Safety criteria paragraph of MIL-E-16400.

3.14.1 Leakage current. Unless otherwise approved by the procuring activity, the leakage current of each equipment of the VFCT terminal shall not exceed 5 mA.

3.15 Processors and software. VFCT terminals containing one or more processors and containing software (including firmware such as ROMs, programmable read-only memories (PROMs), erasable programmable read-only memories (EPROMs), electronically erasable programmable ROMs or electrically alterable ROMs, or nonvolatile RAM E^2 PROMs, programmable logic arrays (PLAs), programmable array logic (PALs) shall conform to the requirements specified in 3.15.1 through 3.15.6. The processors and software requirements specified herein are appplicable only if one or more processors are embedded in the VFCT terminal.

3.15.1 Processors. Processors used in the VFCT terminal shall be subject to procuring activity approval except as specified in a and b:

a. Processors (or computers) that are Navy standard embedded computers
b. Processors (or computers) that are wholly contained on a single integrated circuit, that have a data path of eight bits or less to main memory, that have a maximum addressability of 65,536, or fewer, memory locations, and that are used in hardware intensive applications only (software never to be changed throughout life of the equipment),

3.15.2 Processor programming language. Programming languages for processors used in the VFCT terminal shall be subject to procuring activity approval except as specified in a and b:

a. Navy standard higher order languages, compilers and their support systems, and programming language preprocessors
b. Programming languages used for processors (or computers) that conform to the requirements of 3.15.1b .

3.15.3 Processor programming requirements. The VFCT terminal software program shall conform to the Programming standards and Programming conventions paragraphs of DoD-STD-1679. The program design language (PDL) representation of the software shall be used as a design tool to permit verification of the software performance requirements. The PDL is not required to be executable or compilable.

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3.15.4 Processor programs. VFCT terminal programs shall be contained in firmware (for example, PROMs, EPROMs, E^2PROMs). It shall not be necessary to download software from an external device (for example, a magnetic tape unit) for the VFCT terminal to be able to operate in accordance with the requirements of this specification.

3.15.5 Resource allocation and reserves. Memory storage, I/O channels, and processing time requirements for each program and subprogram shall be determined.

3.15.5.1 Main memory. Main memory for each processor shall have a minimum 25 percent reserve capacity. The reserve capacity shall be measured at peak main memory loading of the processor during processor operational missions. Peak main memory loading shall include all programs and data required for successful operational mission execution. The reserve capacity need not be physically installed in the VFCT terminal, but no cabinet, chassis, or backplane modifications shall be required for reserve capacity installation.

3.15.5.2 Throughput. Throughput for each processor shall have a 25 percent reserve capacity as a minimum. The reserve capacity shall be expressed as a percentage of available capacity at full operational loading over a specific period of time, as determined by operational mission characteristics. In the case of cyclic applications, such as some signal processing applications, this period would be the time between availability of successive input data buffers.

3.15.6 Computer program generation. All processor software shall conform to the requirements of the Software generation paragraph of DoD-STD-1679.

3.16 Identification and marking. Identification and marking shall be in accordance with MIL-E-16400, except as specified in 3.16.1 through 3.16.3.

3.16.1 Weight marking. Equipment weighing in excess of 35 lbs shall have the weight clearly marked on the external surface of the equipment and readily visible during installation and removal. The marking shall be as permanent as the normal life expectancy of the unit to which it is applied.

3.16.2 Electrostatic discharge (ESD) marking. The front and rear panels of each equipment containing ESD components shall be prominently marked with a MIL-STD-129 sensitive electronic devices symbol. An ESD caution, as specified herein, shall be provided on the interior of the equipment enclosure in a location that will be readily visible to personnel prior to gaining access to ESD sensitive parts or assemblies:

CAUTION

THIS EQUIPMENT CONTAINS ASSEMBLIES SENSITIVE TO DAMAGE BY ELECTROSTATIC DISCHARGE (ESD). USE ESD PRECAUTIONARY PROCEDURES WHEN TOUCHING, REMOVING, OR INSERTING ASSEMBLIES.

3.16.3 Color coding and marking. Color coding and marking of pin jacks and other test points for the VFCT terminal shall be in accordance with MIL-STD-415.

3.17 Reliability. The quantitative parameter for the probability that the VFCT terminal will perform its intended function for a specified interval under specified conditions shall be expressed as mean-time-between-failure (MTBF), expressed in continuous hours of operation. The lower test MTBF (θ_1 as defined by MIL-STD-781) for 16-channel VFCT terminals shall be equal to, or greater than, the values specified in TABLE IX.

3.18 Maintainability. The quantitative parameter to identify the probability that a system or component will be restored to a specified operating condition within a given time after a failure is the mean-time-to-repair (MTTR) expressed in minutes of time. The VFCT terminal shall have a MTTR not exceeding 45 minutes and a maximum-corrective-maintenance (M_{maxct}) (95th percentile) not exceeding 60 minutes when corrective maintenance is performed at the organizational level of maintenance by the replacement of assemblies, subassemblies, and chassis-mounted parts. The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction (including a malfunction that allows degraded operation) occurs.

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TABLE IX. VFCT terminal quantitative reliability requirements.

VFCT terminal	Capability	MTBF 1/ (hours)
A	Modulation and demodulation functional mode capability; untwinned and twinned channel capability for both modulation and demodulation functions; Channel No. 1 to Channel No. 16 (untwinned) and Channel No. 1 to Channel No. 8 (twinned) resident channel capability for both modulation and demodulation functions	625
B	Demodulation functional mode; untwinned and twinned channel capability; Channel No. 1 to Channel No. 16 (untwinned) and Channel No. 1 to Channel No. 8 (twinned) resident channel capability	1250

1/ θ_1 as specified in MIL-STD-781

3.19 Workmanship. Workmanship shall be as specified in 3.19.1 and 3.19.2.

3.19.1 General workmanship. General workmanship shall conform to the Workmanship paragraph of MIL-E-16400.

3.19.2 Workmanship screen. The VFCT terminal shall withstand a defect detection vibration screen of random vibration at 0.04 g^2/Hz from 30 Hz to 350 Hz and temperature cycling as specified in 4.9.2.1 and 4.9.2.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

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4.1.4 Test and measuring equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be utilized by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.3)
b. Quality conformance inspection
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Safety verification (see 4.6.12)
d. Inspection of preparation for delivery (see 4.10)

4.3 First article inspection. The number of units required for first article inspection shall be as specified in the contract (see 6.2). First article inspection shall consist of all examination and testing necessary to determine compliance with the reqirements of this specification. First article inspection shall include the tests specified in TABLE X.

TABLE X. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination:						
Weight	3.5.1	4.6.1	X		X	X
Dimensions	3.5.2	4.6.1	X		X	
Parts, materials, and processes	3.4.2.1, 3.4.2.1.4, 3.4.2.4.2, 3.4.2.4.11, 3.4.3.1.3, 3.4.3.3, 3.4.3.3.5, 3.8, 3.8.1, 3.8.1.1, 3.8.1.3 through 3.8.1.7, 3.9.1.4, 3.13.4, 3.15.4	4.6.1	X			
Finish and color	3.8	4.6.1	X	X		
Identification and marking	3.4.1.1, 3.4.2.4.1, 3.4.3.1.1, 3.16 through 3.16.3	4.6.1	X		X	
Safety	3.14	4.6.1	X		X	
Design and construction	3.4.2.1.2, 3.4.2.1.3, 3.4.2.3.1, 3.4.2.3.2, 3.4.2.4.3, 3.4.2.4.11, 3.4.2.5, 3.4.3.1.1, 3.4.3.1.3, 3.4.3.2.1 through 3.4.3.2.5, 3.4.3.3.5, 3.4.3.4, 3.9.1.4, 3.12.1, 3.13.1 through 3.13.5, 3.15.1 through 3.15.6	4.6.1	X			
Workmanship:						
General workmanship	3.19.1	4.9.1	X	X		
Workmanship screen	3.19.2	4.9.2	X	X		
Controls, indicators, and displays	3.4.2.3.1, 3.4.2.3.2, 3.4.2.4.2, 3.4.2.4.3, 3.4.3.2.1 through 3.4.3.2.5, 3.6, 3.6.1	4.6.2	X		X	

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TABLE X. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
VFCT functional tests:						
Modulation function tests:						
Channel designations, digital inputs, and associated binary FSK audio tone frequencies	3.4.1.1	4.6.3.1.1	X			
Logical and signal sense	3.4.1.3	4.6.3.1.2	X			
Digital input, bias distortion, and transition distortion	3.4.1.2, 3.4.2.1.1 through 3.4.2.1.3, 3.4.2.2.1	4.6.3.1.3	X		X	
Tone frequency tolerance and stability	3.4.2.2.2	4.6.3.1.4	X		X	
Tone level and stability	3.4.2.2.3, 3.4.2.4.9	4.6.3.1.5	X		X	
Audio output level adjustment, channel output selection and deselection, and stability	3.4.2.4.2, 3.4.2.4.3	4.6.3.1.6	X		X	
Audio output impedance, balance, and bandwidth	3.4.2.4.4, 3.4.2.4.11	4.6.3.1.7	X		X	
Audio output differential time delay	3.4.2.4.5	4.6.3.1.8	X		X	
Audio output in-band spurious signal levels	3.4.2.4.6, 3.4.2.4.7	4.6.3.1.9	X		X	
Audio output out-of-band spurious signal levels	3.4.2.4.8	4.6.3.1.10	X		X	
Demodulation function tests:						
Channel designations, binary FSK audio tone frequencies, and associated digital outputs	3.4.1.1	4.6.3.2.1	X			
Logical and signal sense	3.4.1.3	4.6.3.2.2	X			
Internally generated spurious signal levels	3.4.3.1.4	4.6.3.2.3	X		X	
Audio input impedance, balance, and bandwidth	3.4.3.1.2, 3.4.3.1.7	4.6.3.2.4	X		X	

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TABLE X. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Audio input differential time delay	3.4.3.1.8	4.6.3.2.5	X		X	
Audio input dynamic range	3.4.3.1.5	4.6.3.2.6	X		X	
Audio input tone frequency tolerance	3.4.3.1.10	4.6.3.2.7	X		X	
Audio input tone level tolerance	3.4.3.1.11	4.6.3.2.8	X		X	
Transmitting bias distortion and isochronous distortion tolerance and digital data output	3.4.3.1.9, 3.4.3.3.2 through 3.4.3.3.4	4.6.3.2.9	X		X	
Audio input time delay compensation	3.4.3.2.5	4.6.3.2.10	X		X	
BER performance for nondiversity operation with no interference	3.4.3.1.5, 3.4.3.2.1, 3.4.3.3.6	4.6.3.2.11	X	X		
BER performance for nondiversity operation in AWGN	3.4.3.1.6, 3.4.3.2.1, 3.4.3.3.7	4.6.3.2.12	X		X	
BER performance for tone diversity operation in AWGN	3.4.3.1.6, 3.4.3.2.2, 3.4.3.3.8	4.6.3.2.13	X		X	
BER performance for audio input diversity operation in AWGN	3.4.3.1.6, 3.4.3.2.3, 3.4.3.3.9	4.6.3.2.14	X		X	
BER performance for tone plus audio input diversity operation in AWGN	3.4.3.1.6, 3.4.3.2.4 3.4.3.3.10	4.6.3.2.15	X		X	
Back-to-back modulation and demodulation function tests:						
BER performance for nondiversity operation with no interference	3.4.3.1.5, 3.4.2.3.1, 3.4.3.2.1, 3.4.3.3.6, 3.4.4.1	4.6.3.3.1	X	X		
BER performance for nondiversity operation in AWGN	3.4.2.3.1, 3.4.3.1.6, 3.4.3.2.1, 3.4.3.3.7, 3.4.4.2	4.6.3.3.2	X		X	
BER performance for tone diversity operation in AWGN	3.4.3.1.6, 3.4.2.3.2, 3.4.3.2.2, 3.4.3.3.8, 3.4.4.3	4.6.3.3.3	X		X	
Digital data output distortion	3.4.2.2.1, 3.4.2.3.1, 3.4.3.1.5, 3.4.3.1.9, 3.4.3.2.1, 3.4.3.3.2, 3.4.4.4	4.6.3.3.4	X	X		
Crosstalk	3.4.2.3.1, 3.4.3.1.6, 3.4.3.2.1, 3.4.4.5	4.6.3.3.5	X		X	

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TABLE X. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Operating tests	3.4	4.6.4		X		
Software program performance tests	3.15.1 through 3.15.6	4.6.13	X			
Electrical tests:						
Steady-state voltage and frequency	3.9.1.1	4.6.7.1	X		X	
Transient voltage	3.9.1.1	4.6.7.2	X		X	
Transient frequency	3.9.1.1	4.6.7.3	X		X	
Power interruption	3.9.1.1, 3.9.1.5	4.6.7.4	X		X	
Power and power factor	3.9.1.1, 3.9.1.2	4.6.7.5	X		X	
Leakage current	3.14.1	4.6.12.1	X		X	
Airborne noise	3.12.3	4.6.10	X			
Structureborne noise	3.12.3	4.6.11	X			
Enclosure	3.12.2	4.6.9	X			
EMC	3.10	4.6.8	X			
Environmental:						
Low temperature	3.7.1	4.6.5.1	X			X
High temperature	3.7.1	4.6.5.2	X			X
Humidity	3.7.2	4.6.5.3	X			X
Salt fog	3.7.3	4.6.5.4	X		X	
Shock	3.7.4	4.6.5.5	X			
Vibration	3.7.5	4.6.5.6	X			
Inclination	3.7.6	4.6.5.7	X			
Interchangeability	3.8.1.2	4.6.6	X			
Reliability	3.4.1.4, 3.17	4.7	X			
Maintainability	3.18, 3.11 through 3.11.7	4.8	X			

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be made on each VFCT terminal offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in TABLE X, Group A.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. Production control inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests to determine temperature rise produced in operation and ability of the VFCT terminal to withstand this heat; tests of efficiency; and tests of performance with other VFCT terminals in a back-to-back configuration. The tests shall be performed on the complete VFCT terminal as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE X, Group B.

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4.4.2.1 Sampling plan for production control inspection (Group B). VFCT terminals submitted for production control inspection shall be selected by the contractor under the supervision of the quality assurance representative (QAR) and shall be representative of the current production. Production control inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Production control inspection shall be performed on inspection lots that have passed production inspection. The acceptable quality level shall be a 6.5 percent defective for each of the Group B examinations and tests and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective VFCT terminals and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be accomplished on a sampling basis as specified in the contract (see 6.2). Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units and the VFCT terminal as a whole; tests of the effects of changes of environment (such as extremes of temperature and humidity); and tests of the effects of shock and vibration. Environmental inspection shall include the examinations and tests specified in TABLE X, Group C.

4.4.3.1 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of the failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming production control and environmental sample VFCT terminals. Unless otherwise specified (see 6.2), sample VFCT terminals which have been subjected to and passed, production control and environmental inspections, or both, may be accepted on the contract provided all damage has been repaired and the VFCT terminals are resubjected to, and pass, production inspection.

4.5 Testing criteria. Unless otherwise specified herein, examinations and tests shall be in accordance with the quality assurance provisions of MIL-E-16400 specified in a through c:

a. Test facilities, chambers, and apparatus
b. Standard test conditions
c. Performance of test

4.6 Test methods. Test methods shall be as specified in 4.6.1 through 4.6.13.

4.6.1 Surface examination. The VFCT terminal shall be examined to ensure compliance with the requirements specified in a through g:

a. Weight (see 3.5.1)
b. Dimensions (see 3.5.2)
c. Parts, materials, and processes (see 3.8)
d. Finish and color (see 3.8)
e. Design and construction (see 3.12)
f. Safety (see 3.14)
g. Identification and marking (see 3.16)

In the examination process, the VFCT terminal shall not be disassembled and reassembled in a manner that will affect the performance, durability, or appearance of the VFCT terminal.

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4.6.2 Controls, indicators, and displays test. Each of the VFCT terminal controls, indicators, and displays shall be subjected to complete and comprehensive testing to verify correct and specified operation and functioning and to verify that these items cannot be made to malfunction. Testing shall include, but shall not be limited to:

a. Testing of out of sequence control operation
b. Testing of operation of more than one control simultaneously
c. Testing to determine the impact of control actuation on the VFCT terminal in the operating mode
d. Testing to determine the accuracy and response of indicators and displays
e. Testing to determine accuracy of human engineering design

4.6.3 VFCT functional tests. The VFCT terminal shall be subjected to tests to determine compliance with the VFCT requirements specified in 3.4. Unless otherwise specified in 4.6.3.1 through 4.6.3.3.5, binary FSK tone levels shall be as specified in a and b:

a. For all modulation channel tones outputted from the VFCT terminal, tone levels shall be nominally equal in level (for example, the tone levels of one modulation channel shall be no greater in level nor no lower in level than the tone levels of the other modulation channels).
b. For all demodulation channel tones inputted to the VFCT terminal, the tones shall be nominally equal in level (for example, the tone levels of one demodulation channel shall be no greater in level nor no lower in level than the tone levels of the other demodulation channels).

4.6.3.1 Modulation function tests. The modulation function of the VFCT terminal shall be subjected to the modulation function tests specified in 4.6.3.1.1 through 4.6.3.1.10.

4.6.3.1.1 Channel designations, digital inputs, and associated binary FSK audio tone frequencies tests. The modulation function of the VFCT terminal, with the composite VFCT baseband audio output of the VFCT terminal terminated in a 600-ohm resistive load impedance, shall be subjected to tests to verify that the binary FSK tones that are present at the composite VFCT baseband audio output of the VFCT terminal for each modulation channel are being modulated by the associated modulation channel digital data input as specified in TABLE I for nondiversity operation and as specified in TABLE II for tone diversity operation (for example, the digital data input for modulation Channel No. 1. modulates:

a. The binary FSK tone pair specified for Channel No. 1 in TABLE I for nondiversity operation
b. Both the binary FSK tone pairs specified for Channel No. 1 in TABLE II for tone diversity operation)

The tests specified herein shall be conducted, in turn, for low level digital data inputs and for high level digital data inputs to each modulation channel.

4.6.3.1.2 Modulation function logical and signal sense tests. The modulation channels of the VFCT terminal set up, in turn, for standard and reverse sense operation, shall be subjected to tests to verify modulation function logical and signal sense. When digital data of the mark and space logical sense specified in 3.4.1.3 are inputted to each modulation channel for standard sense operation, the mark and space signal sense of the associated modulation channel binary FSK tones at the composite VFCT baseband audio output of the VFCT terminal shall be determined and shall correspond in signal sense to the mark and space signal sense for the modulation channel specified in TABLE I for nondiversity operation and in TABLE II for tone diversity operation (for example, for nondiversity operation, a mark logical signal (as specified in 3.4.1.3) inputted to the digital data input for the modulation channel shall cause a mark binary FSK tone (lower frequency tone of the binary FSK tone pair for the channel) to be outputted from the composite VFCT baseband audio output of the VFCT terminal for the modulation channel). When digital data of the mark and space logical sense specified in 3.4.1.3 are inputted to each modulation channel for reverse sense operation, the mark and space signal sense of the associated modulation channel binary FSK tones at the composite VFCT baseband audio output of the VFCT terminal shall be determined and shall be opposite in mark and space signal sense to the signal sense specified in TABLE I for nondiversity operation and in TABLE II for tone diversity operation (for example, for nondiversity operation, a mark logical sense signal (as specified in 3.4.1.3) inputted to the digital data input for the modulation channel shall cause the upper frequency tone of the binary FSK tone pair (the space FSK signal state as specified in 3.4.1.3) for the channel to be outputted from the composite VFCT baseband audio

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output of the VFCT terminal for the modulation channel). The tests specified herein shall be conducted, in turn, for low level digital data inputs and for high level digital data inputs to each modulation channel. The composite VFCT baseband audio output of the VFCT terminal shall be terminated in a 600-ohm resistive load impedance for these tests.

4.6.3.1.3 Digital data input, bias distortion, and transition distortion tests. Transmitting isochronous distortion, transmitting bias distortion and transition distortion shall be determined for each modulation channel of the VFCT terminal for nondiversity modulation function operation. The composite VFCT baseband audio output of the VFCT terminal shall be terminated in a 600-ohm resistive load impedance for these tests. For these tests the digital data inputs to the modulation channel shall be of the types specified in a through e:

a. Type A: Undistorted (isochronous distortion less than or equal to 0.1 percent)
b. Type B: Marking bias distortion of 2 percent (average and peak)
c. Type C: Spacing bias distortion of 2 percent (average and peak)
d. Type D: Marking bias distortion of 35 percent (average and peak)
e. Type E: Spacing bias distortion of 35 percent (average and peak)

For these tests the digital data inputs shall be applied to the modulation channel digital data input interfaces with associated interface conditions as specified in f through k:

f. Condition 1: Low level interface, balanced input
g. Condition 2: Low level interface, unbalanced input
h. Condition 3: High level interface, 60 mA positive loop current
i. Condition 4: High level interface, 60 mA negative loop current
j. Condition 5: High level interface, 120 mA positive loop current
k. Condition 6: High level interface, 120 mA negative loop current

For each of the digital data input types specified in a through e, applied under each of the digital data interface conditions specified in f through k, peak and average values for each of the times specified in l through o, as a minimum, shall be determined:

l. Time duration for the mark tones
m. Time duration for the space tones
n. Mark tone to space tone transition time
o. Space tone to mark tone transition time

For each digital data interface condition specified in f through k, the transmitting isochronous distortion and the transition distortion for the Type A digital data input to each modulation channel shall be determined. For each of the digital data interface conditions specified in f through k, the transmitting bias distortion and the transition distortion for the Type B, Type C, Type D, and Type E digital data inputs to each modulation channel shall be determined.

4.6.3.1.4 Tone frequency tolerance and stability tests. The VFCT terminal shall be inserted into a temperature controlled chamber for these tests, and measurements shall be made at ambient temperatures inside the chamber in the range of from 0°C to 50°C at 10°C increments (that is, at 0°C, 10°C, 20°C, 30°C, 40°C, and 50°C) after the temperature of the VFCT terminal has stabilized at each measurement temperature. The frequency of each mark and space tone at the composite VFCT baseband audio output of the VFCT terminal (with the output terminated in a 600-ohm resistive load impedance) for each modulation channel of the VFCT terminal in the nondiversity mode of operation shall be determined at each specified test temperature. For the purposes of these tests, output of a single mark or space tone at a time at the composite VFCT baseband audio output of the VFCT terminal is acceptable.

4.6.3.1.5 Tone level tolerance and stability tests. For these tests the composite VFCT baseband audio output of the VFCT terminal shall be terminated in a 600-ohm resistive load impedance. With the modulation function of the VFCT terminal operating in the nondiversity mode, each modulation channel shall be placed in a constant marking condition and the mark tone voltage level across the 600-ohm resistive load impedance shall be measured for each modulation channel after the composite VFCT baseband audio output of the VFCT terminal has been adjusted to a fixed average power level. With the modulation function of the VFCT terminal still operating in the nondiversity mode and with the composite VFCT baseband audio output of the VFCT terminal at the same fixed average power level, each modulation channel shall be placed in a constant spacing condition and the space tone voltage level across the 600-ohm resistive load impedance

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shall be measured for each modulation channel. These measurements of mark and space tone levels at the composite VFCT baseband audio output of the VFCT terminal shall be made, in turn, at fixed composite VFCT baseband audio output average power levels of -13 dBm, 0 dBm, and +3 dBm (with respect to the 600-ohm resistive load impedance). At each fixed composite VFCT baseband audio output average power level the voltage ratio in dB shall be determined for:

a. The mark tone to space tone voltage levels (or the space tone to mark tone voltage levels as appropriate) for each modulation channel
b. The voltage level of the mark or space tone having the greatest voltage level to the voltage level of the mark or space tone having the lowest voltage level.

4.6.3.1.6 Audio output level adjustment, channel output selection and deselection, and stability. For these tests the composite VFCT baseband audio output of the VFCT terminal shall be terminated in a 600-ohm resistive load impedance, the modulation function of the VFCT terminal shall be operating in the nondiversity mode, and random mark and space digital data (uncorrelated in bit transitions between modulation channels) shall be inputted to each of the modulation channel digital data inputs. The capability of the VFCT terminal composite VFCT baseband audio output to be adjusted to each dBm average power level (with respect to the 600-ohm resistive load impedance) from -24 dBm to +3 dBm, inclusive, shall be verified. The composite VFCT baseband audio output from the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm, 0 dBm, and +3 dBm (with respect to the 600-ohm resistive load impedance); at each fixed average power level the composite VFCT baseband audio output average power level into the 600-ohm resistive load shall be monitored over a 60-minute time period, as a minimum, to establish the range of output level variation about the set value. The range of variation of the composite VFCT baseband audio output average power (into the 600-ohm resistive load impedance) about a set average power output level and as a function of the number of modulation channel binary FSK tones present in the composite output signal shall be determined. For this test the composite VFCT baseband audio output from the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm, 0 dBm, and +3 dBm (with respect to the 600-ohm resistive load impedance) in each of two test conditions. For the first test condition, the fixed average power level shall be set with the binary FSK tones of all modulation channels outputted from the VFCT terminal into the 600-ohm resistive load impedance, and the range of variation of the composite VFCT baseband audio output average power about the set level determined; then, one-by-one the modulation channels and their associated binary FSK tones shall be deselected. After deselection of each modulation channel, the level of the composite VFCT baseband audio output average power into the 600-ohm resistive load impedance and the variation of the average power about that level shall be determined. For the second test condition, the fixed average power level shall be set with the binary FSK tones of only one of the modulation channels outputted from the VFCT terminal into the 600-ohm resistive load impedance, and the range of variation of the average power of the binary FSK tones for this channel about the set level determined; one-by-one the modulation channels and their associated binary FSK tones shall be selected and outputted from the VFCT terminal. After selection and output of each additional modulation channel the level of the composite VFCT baseband audio output average power into the 600-ohm resistive load impedance and the variation of the average power about that level shall be determined.

4.6.3.1.7 Audio output impedance, balance, and bandwidth tests. The output impedance, return loss, and longitudinal current suppression (with respect to a 600-ohm resistive load impedance terminating the output) of the composite VFCT baseband audio output A of the VFCT terminal shall be determined over the minimum frequency range of 350 Hz to 3050 Hz.

4.6.3.1.8 Audio output differential time delay tests. The differential time delay (over the minimum frequency range of from 350 Hz to 3050 Hz at the binary FSK tone frequencies) through the quasi-analog circuitry of the modulation function of the VFCT terminal from the point in the circuitry at which each modulation channel binary FSK tone is initially present to the composite VFCT baseband audio output connector of the VFCT terminal, shall be determined.

4.6.3.1.9 Audio output in-band spurious signal levels tests. For these tests the composite VFCT baseband audio output of the VFCT terminal shall be terminated in a 600-ohm reistive load impedance, the modulation function of the VFCT terminal shall be operating in the nondiversity mode, and the digital data input to each of the modulation channel digital data inputs shall be, in turn, constant marking and constant spacing (although the inputs to each of the modulation channels will not necessarily be the same for each series of measurements). In-band spurious signal levels shall be determined, in turn, at fixed composite VFCT baseband audio output average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm resistive load impedance). With 16 constant frequency (unmodulated) tones (one for each of the 16 modulation channel nondiversity binary FSK tone pairs) present at the VFCT terminal composite

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VFCT baseband audio output, the audio output level shall be set to the fixed composite VFCT baseband audio output level. One modulation channel output shall be deselected, and the frequency and signal level (in a 1-Hz bandwidth) of all spurious signals (signals at frequencies other than the remaining 15 modulation channel tones present and including harmonics and IM products) at the composite VFCT baseband audio output of the VFCT terminal in the frequency range from 1 Hz to 4125 Hz shall be determined. The constant marking or constant spacing inputs to one or more of the digital data inputs of the modulation channels shall be reversed (for example, constant marking to constant spacing, constant spacing to constant marking) and the preceding measurements repeated. The specified measurement process shall be repeated for a representative number of combinations of constant marking and constant spacing tones for the modulation channels. The modulation channel output that was deselected shall then be restored and another of the modulation channel outputs shall be deselected. The determination of spurious signal frequencies and levels in the 1-Hz to 4125-Hz frequency range for combinations of 15 modulation channel constant marking and constant spacing tones shall be repeated. Spurious signal frequencies and levels in the 1-Hz to 4125-Hz frequency range shall be determined, in turn, when each of the 16 modulation channel outputs is deselected with the remaining 15 modulation channel outputs present.

4.6.3.1.10 Audio output out-of-band spurious signal levels tests. For these tests the composite VFCT baseband audio output of the VFCT terminal shall be terminated in a 600-ohm resistive load impedance, the modulation function of the VFCT terminal shall be operating in the nondiversity mode, and the digital data input to each of the modulation channel digital data inputs shall be constant marking. Out-of-band spurious signal levels shall be determined, in turn, at fixed composite VFCT baseband audio output average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm resistive load impedance). The frequency and signal level (in a 1-Hz bandwidth) of all spurious signals (for example, harmonics, IM products) present at the composite VFCT baseband audio output of the VFCT terminal in the frequency range of from 4125 Hz to 30 MHz shall be determined.

4.6.3.2 Demodulation function tests. The demodulation function of the VFCT terminal shall be subjected to the demodulation function tests specified in 4.6.3.2.1 through 4.6.3.2.15.

4.6.3.2.1 Channel designations, binary FSK audio tone frequencies, and associated digital outputs tests. The demodulation function of the VFCT terminal shall be subjected to tests to verify that the digital data output from each demodulation channel results from demodulation of the associated demodulation channel binary FSK tone pairs as specified in TABLE I for nondiversity operation and as specified in TABLE II for tone diversity operation (for example, the digital data output from demodulation Channel No. 1 shall result from:

a. Demodulation of the binary FSK tone pair for Channel No. 1 as specified in TABLE I for nondiversity operation
b. Demodulation (and combining) of both (and, in turn, each of) the binary FSK tone pairs for Channel No. 1 as specified in TABLE II for tone-diversity operation

Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested by applying binary FSK modulated signals to the input. Each digital data output (low level, high level) of each demodulation channel shall be tested in turn.

4.6.3.2.2 Demodulation function logical and signal sense. The demodulation channels of the VFCT terminal set up, in turn, for standard and reverse sense operation, shall be subjected to tests to verify demodulation function logical and signal sense. For standard sense operation, demodulation channel mark and space binary FSK signals of the signal sense specified in 3.4.1.3 and, in turn, corresponding to the signal sense specified in TABLE I for nondiversity operation and in TABLE II for tone diversity operation, shall be applied to the composite VFCT baseband audio input of the VFCT terminal and the logical sense of the resulting demodulation channel digital data output shall be determined and shall correspond to the mark and space logical sense as specified in 3.4.1.3 (for example, for nondiversity operation, a mark binary FSK tone (lower frequency tone of the binary FSK tone pair) for the demodulation channel shall cause a mark logical sense digital signal (as specified in 3.4.1.3) to be outputted from the digital data output of the demodulation channel). For reverse sense operation, demodulation channel mark and space binary FSK signals of the signal sense specified in 3.4.1.3 and, in turn, corresponding to the signal sense specified in TABLE I for nondiversity operation and in TABLE II for tone diversity operation shall be applied to the composite VFCT baseband audio input of the terminal and the logical sense of the resulting demodulation channel digital output shall be determined and shall correspond to the opposite mark and space logical sense as specified in 3.4.1.3 (for example, for nondiversity operation, a mark binary FSK tone (lower frequency tone of the binary FSK tone pair) for the demodulation channel shall cause a space logical signal (as specified in 3.4.1.3) to be outputted from the digital data output of the demodulation channel). Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested. Each digital data output (low level, high level) of each demodulation channel shall be tested in turn.

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4.6.3.2.3 Internally generated spurious signal levels tests. For these tests the composite VFCT baseband audio input of the VFCT terminal shall be terminated in a 600-ohm resistive load impedance, and no audio signals shall be inputted to the composite VFCT baseband audio input of the VFCT terminal. The frequency and signal level (in a 1-Hz bandwidth) of all signals present at the composite VFCT baseband audio input of the VFCT terminal in the frequency range of from 1 Hz to 30 MHz with the VFCT terminal demodulation function (and the modulation function (if present) at an output level of +3 dBm into a 600-ohm resistive load with the maximum number of modulation channels output) enabled shall be determined. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

4.6.3.2.4 Audio input impedance, balance, and bandwidth tests. The input impedance, return loss, and longitudinal current suppression (with respect to a 600-ohm resistive source impedance) of the composite VFCT baseband audio input of the VFCT terminal shall be determined over the minimum frequency range of 350 Hz to 3050 Hz. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

4.6.3.2.5 Audio input differential time delay tests. The differential time delay (over the minimum frequency range of from 350 Hz to 3050 Hz at the binary FSK tone frequencies) through the quasi-analog circuitry of the demodulation function of the VFCT terminal from the composite VFCT baseband audio input connector of the VFCT terminal to the point in the circuitry at which th binary FSK tone is coverted to another form (for example, input to VFCT terminal demodulation function: limiter, analog to digital converter), shall be determined. For these tests any adjustable time delay compensation shall be adjusted for minimum time delay (see 3.4.3.2.5). Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

4.6.3.2.6 Audio input dynamic range tests. For these tests the demodulation function of the VFCT terminal shall be operating in the nondiversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the 16 demodulation channels shall be inputted to the composite VFCT baseband audio input of the VFCT terminal (without AWGN or other interference) by a 600-ohm resistive impedance signal source. The composite VFCT baseband audio input to the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedance of the VFCT terminal); at each fixed average power level the BER of digital data output from the demodulation channels of the VFCT terminal shall be determined. The composite VFCT baseband audio input to the VFCT terminal shall be increased in input level beyond the 0 dBm level (with respect to the 600-ohm input impedance of the VFCT terminal) and BER measurements shall be made to determine the composite VFCT baseband audio input levels at which the BER has increased 10 times and 100 times over the BER determined at the 0 dBm average power input level. The composite VFCT baseband audio input level shall be decreased in level below the -13 dBm level (with respect to the 600-ohm input impedance of the VFCT terminal) and BER measurements shall be made to determine the composite VFCT baseband audio input levels at which the BER has increased 10 times and 100 times over the BER determined at the -13 dBm average power input level. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

4.6.3.2.7 Audio input tone frequency tolerance tests. For these tests the demodulation function of the VFCT terminal shall be operating in the nondiversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the 16 demodulation channels shall be inputted to the composite VFCT baseband audio input of the VFCT terminal (without AWGN or other interference) by a 600-ohm resistive impedance signal source. The composite VFCT baseband audio input to the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedance of the VFCT terminal): at each fixed average power level the BER of digital data outputted from the demodulation channels of the VFCT terminal shall be determined for the binary FSK modulated tone demodulation channel inputs to the VFCT terminal as specified in a through d:

a. Binary FSK modulated tones in accordance with TABLE I except that each mark and space tone frequency shall be increased by 3 Hz

b. Binary FSK modulated tones in accordance with TABLE I except that each mark and space tone frequency shall be decreased by 3 Hz

c. Binary FSK modulated tones in accordance with TABLE I except that each mark tone frequency shall be increased by 3 Hz and each space tone frequency shall be decreased by 3 Hz

d. Binary FSK modulated tones in accordance with TABLE I except that each mark tone frequency shall be decreased by 3 Hz and each space tone frequency shall be increased by 3 Hz

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The BER measurement results specified herein shall be compared with the BER measurement results of 4.6.3.2.6 at the same composite VFCT baseband audio input average power level. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

4.6.3.2.8 Audio input tone level tolerance tests. For these tests the demodulation function of the VFCT terminal shall be operating in the nondiversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the 16 demodulation channels shall be inputted to the composite VFCT baseband audio input of the VFCT terminal (without AWGN or other interference) by a 600-ohm resistive impedance signal source. The composite VFCT baseband audio input to the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedance of the VFCT terminal); at each fixed average power level the BER of digital data outputted from the demodulation channels of the VFCT terminal shall be determined for the binary FSK modulated tone demodulation channel inputs to the VFCT terminal as specified in a and b:

a. The mark tone for each demodulation channel 2.0 dB lower in level than the space tone for the demodulation channel
b. The space tone for each demodulation channel 2.0 dB lower in level than the mark tone for the demodulation channel

The BER measurement results specified herein shall be compared with the BER measurement results of 4.6.3.2.6 at the same composite VFCT baseband audio input average power level. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

4.6.3.2.9 Transmitting bias distortion and isochronous distortion tolerance, and digital data output tests. These tests shall be conducted on each demodulation channel of the VFCT terminal. For these tests the demodulation function of the VFCT terminal shall be operating in the nondiversity mode, binary FSK modulated composite VFCT baseband audio signals for each of the 16 demodulation channels shall be inputted to the composite VFCT baseband audio input of the VFCT terminal (without AWGN or other interference) by a 600-ohm resistive impedance signal source, and the transition distortion shall be less than or equal to 5 percent. The composite VFCT baseband audio input to the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedance of the VFCT terminal); the tests specified in a through e shall be conducted at each fixed average power level. For these tests the binary FSK modulated tone inputs to the demodulation channel shall be of the types specified in a through e:

a. Type A: Undistorted (isochronous distortion less than or equal to 0.1 percent)
b. Type B: Marking transmitting bias distortion of 2 percent (average and peak)
c. Type C: Spacing transmitting bias distortion of 2 percent (average and peak)
d. Type D: Marking transmitting bias distortion of 35 percent (verage and peak)
e. Type E: Spacing transmitting bias distortion of 35 percent (average and peak)

For these tests measurements shall be made at the digital data outputs of the demodulation channels under the interface conditions specified in f through k:

f. Condition 1: Low level interface, balanced output
g. Condition 2: Low level interface, unbalanced output
h. Condition 3: High level interface, 60 mA positive loop current
i. Condition 4: High level interface, 60 mA negative loop current
j. Condition 5: High level interface, 120 mA positive loop current
k. Condition 6: High level interface, 120 mA negative loop current

For each of the binary FSK modulated tone input types specified in a through e and for each of the digital data interface conditions specified in f through k, the information specified in l and m shall be determined for the digital data output from each demodulation channel:

l. Peak and average bias distortion
m. BER

If the BER for any of the tests exceeds the BER determined for Type A, Condition 1 by a factor equal to or greater than 10 additional tests shall be conducted to determine the level and type of transmitting bias distortion at the input to the VFCT terminal at which the BER has increased to 10 times the BER experienced for the Type A, Condition 1 measurement. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

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4.6.3.2.10 Audio input time delay compensation tests. For these tests the demodulation function of the VFCT terminal shall be operating in the audio input diversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the 16 demodulation channels plus AWGN shall be inputted identically to each of the composite VFCT baseband audio inputs (A and B) of the VFCT terminal by a 600-ohm resistive impedance signal source such that the signal levels and SNDR will be identical at each VFCT terminal composite VFCT baseband audio input (A and B). The composite VFCT baseband audio signals to both inputs (A and B) to the VFCT terminal shall each be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedances of the VFCT terminal); at each fixed average power level, BER (in the minimum range of 1×10^{-4} to 1×10^{-2}) versus SNDR measurements of the digital data output from the demodulation channels of the VFCT terminal shall be made for the input conditions specified in a and b:

a. A time delay of 0 ms between the composite VFCT baseband audio signals applied to VFCT terminal audio input A and the composite VFCT baseband audio signals applied to VFCT terminal audio input B

b. A time delay of 3.5 ms between the composite VFCT baseband audio signals applied to VFCT terminal audio input A and the composite VFCT baseband audio signals applied to VFCT terminal audio input B.

4.6.3.2.11 Demodulation function BER performance tests for nondiversity operation with no external interference. For these tests the demodulation function of the VFCT terminal shall be operating in the nondiversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the 16 demodulation channels shall be inputted to the composite VFCT baseband audio input of the VFCT terminal (without AWGN or interference) by a 600-ohm resistive impedance signal source. The composite VFCT baseband audio input to the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm, 0 dBm, and +3 dBm (with respect to the 600-ohm input impedance of the VFCT terminal); at each fixed average power level the digital data outputs of each of the demodulation channels shall be monitored for bit errors for a minimum continuous period of one hour.

4.6.3.2.12 Demodulation function BER performance tests for nondiversity operation in AWGN. For these tests the demodulation function of the VFCT terminal shall be operating in the nondiversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the 16 demodulation channels plus AWGN shall be inputted to the composite VFCT baseband audio input of the VFCT terminal by a 600-ohm resistive impedance signal source. The composite VFCT baseband audio signals inputted to the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedance of the VFCT terminal); at each fixed average power level, BER (in the minimum range from 1×10^{-4} to 1×10^{-2}) versus SNDR measurements of the digital data output from the demodulation channels of the VFCT terminal shall be made. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

4.6.3.2.13 Demodulation function BER performance tests for tone-diversity operation in AWGN. For these tests the demodulation function of the VFCT terminal shall be operating in the tone-diversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the eight demodulation channels (two binary FSK modulated tone pairs per demodulation channel) plus AWGN shall be inputted to the composite VFCT baseband audio input of the VFCT terminal by a 600-ohm resistive impedance signal source. Composite VFCT baseband audio signals of nominally equal tone levels (16 binary FSK modulated tones) inputted to the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedance of the VFCT terminal); at each fixed average power level, BER (in the minimum range of from 1×10^{-4} to 1×10^{-2}) versus SNDR measurements of the digital data output from the demodulation channels of the VFCT terminal shall be made for the inputted conditions specified in a through c:

a. The individual levels of the lower frequency binary FSK tone pairs of the demodulation channels equal to the individual levels of the upper frequency binary FSK tone pairs of the demodulation channels

b. The individual levels of the lower frequency binary FSK tone pairs of the demodulation channels each 5.5 dB lower in level than the individual levels of the upper frequency binary FSK tone pairs of the demodulation channels

c. The individual levels of the upper frequency binary FSK tone pairs of the demodulation channels each 5.5 dB lower in level than the individual levels of the lower frequency binary FSK tone pairs of the demodulation channels.

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Once AWGN input levels have been set to establish the measurement SNDR for input condition a herein, these noise levels shall be maintained during the lowering of the signal levels of certain tones for input conditions b and c. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested.

4.6.3.2.14 Demodulation function BER performance tests for audio input diversity operation in AWGN. For these tests the demodulation function of the VFCT terminal shall be operating in the audio input diversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the 16 demodulation channels plus AWGN shall be inputted as specified herein to each of the composite VFCT baseband audio inputs (A and B) of the VFCT terminal by a 600-ohm resistive impedance signal source. Composite VFCT baseband audio signals of nominally equal tone levels for the 16 demodulation channels inputted to each of the composite VFCT baseband audio inputs (A and B) of the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedances of the VFCT terminal); at each fixed average power level, BER (in the minimum range of from 1×10^{-4} to 1×10^{-2}) versus SNDR measurements of the digital data output from the 16 demodulation channels of the VFCT terminal shall be made for the input conditions specified in a through c:

a. The individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input A of the VFCT terminal equal to the individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input B of the VFCT terminal

b. The individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input A of the VFCT terminal each 5.5 dB lower in level than the individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input B of the VFCT terminal

c. The individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input B of the VFCT terminal each 5.5 dB lower in level than the individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input A of the VFCT terminal.

Once AWGN input levels have been set to establish the measurement SNDR for input condition a herein, these noise levels shall be maintained during the lowering of the signal levels of certain tones for input conditions b and c.

4.6.3.2.15 Demodulation function BER performance tests for tone plus audio input diversity operation in AWGN. For these tests the demodulation function of the VFCT terminal shall be operating in the tone plus audio input diversity mode, and binary FSK modulated composite VFCT baseband audio signals for each of the eight demodulation channels (four binary FSK modulated tone pairs total per demodulation channel) plus AWGN shall be inputted as specified herein to each of the composite VFCT baseband audio inputs (A and B) of the VFCT terminal by a 600-ohm resistive impedance signal source. Composite VFCT baseband audio signals of nominally equal tone levels for the eight demodulation channels (16 binary FSK modulated tone pairs) inputted to each of the composite VFCT baseband audio inputs (A and B) of the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and 0 dBm (with respect to the 600-ohm input impedances of the VFCT terminal); at each fixed average power level, BER (in the minimum range of from 1×10^{-4} to 1×10^{-2}) versus SNDR measurements of the digital data outputted from the eight demodulation channels of the VFCT terminal shall be made for the input conditions specified in a through e:

a. The individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input A of the VFCT terminal equal to the individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input B of the VFCT terminal (equal levels for the four binary FSK modulated tone pairs of each demodulation channel)

b. The individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input A of the VFCT terminal and the individual levels of the eight lower frequency binary FSK tone pairs of the demodulation channels inputted to composite VFCT baseband audio input B of the VFCT terminal each 5.5 dB lower in level than the individual levels of the eight upper frequency binary FSK tone pairs of the demodulation channels inputted to composite VFCT baseband audio input B of the VFCT terminal

c. The individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input A of the VFCT terminal and the individual levels of the eight upper frequency binary FSK tone pairs of the demodulation channels inputted to composite VFCT baseband audio input B of the VFCT terminal each 5.5 dB lower in level than the individual levels of the eight lower frequency binary FSK tone pairs of the demodulation channels inputted to composite VFCT baseband audio input B of the VFCT terminal

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d. The individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input B of the VFCT terminal and the individual levels of the eight lower frequency binary FSK tone pairs of the demodulation channels inputted to composite VFCT baseband audio input A of the VFCT terminal each 5.5 dB lower in level than the individual levels of the eight upper frequency binary FSK tone pairs of the demodulation channels inputted to composite VFCT baseband audio input A of the VFCT terminal

e. The individual levels of the 16 binary FSK modulated tone pairs inputted to composite VFCT baseband audio input B of the VFCT terminal and the individual levels of the eight upper frequency binary FSK tone pairs of the demodulation channels inputted to composite VFCT baseband audio input A of the VFCT terminal each 5.5 dB lower in level than the individual levels of the eight lower frequency binary FSK tone pairs of the demodulation channels inputted to composite VFCT baseband audio input A of the VFCT terminal.

Once AWGN input levels have been set to establish the measurement SNDR for input condition a herein, these noise levels shall be maintained during the lowering of the signal levels of certain tones for input conditions b, c, d, and e.

4.6.3.3 Modulation and demodulation function tests. The modulation and demodulation functions of the VFCT terminal shall be subjected to the modulation and demodulation function tests specified in 4.6.3.3.1 through 4.6.3.3.5.

4.6.3.3.1 Modulation and demodulation function BER performance tests for nondiversity operation with no external interference. The VFCT terminal shall be subjected to the demodulation function BER performance tests for nondiversity operation with no external interference as specified in 4.6.3.2.11, except that the modulation function of the VFCT terminal shall be connected back-to-back to the demodulation function of the VFCT terminal at the composite VFCT baseband audio output and audio input.

4.6.3.3.2 Modulation and demodulation function BER performance tests for nondiversity operation in AWGN. The VFCT terminal shall be subjected to the demodulation function BER performance tests for nondiversity operation in AWGN as specified in 4.6.3.2.12, except that the modulation function of the VFCT terminal shall be connected back-to-back to the demodulation function of the VFCT terminal at the composite VFCT baseband audio output and audio input.

4.6.3.3.3 Modulation and demodulation function BER performance tests for tone diversity operation in AWGN. The VFCT terminal shall be subjected to the demodulation function BER performance tests for tone diversity operation in AWGN as specified in 4.6.3.2.13, except that the modulation function of the VFCT terminal shall be connected back-to-back to the demodulation function of the VFCT terminal at the composite VFCT baseband audio output and audio input.

4.6.3.3.4 Modulation and demodulation function digital data output distortion tests. For these tests the VFCT terminal shall be operating in the nondiversity mode and the modulation function of the VFCT terminal shall be connected back-to-back to the demodulation function of the VFCT terminal at the composite VFCT baseband audio output and audio input. The composite VFCT baseband audio input to the composite VFCT baseband audio input of the VFCT terminal shall be adjusted, in turn, to fixed average power levels of -13 dBm and +3 dBm (with respect to the 600-ohm input impedance of the VFCT terminal) for the measurements specified herein. With 16 undistorted (isochronous and telegraph distortion less than 0.1 percent) random mark and space digital data inputs applied to the modulation channel digital data inputs of the modulation function of the VFCT terminal, the peak and average isochronous distortion and the peak and average telegraph distortion of the digital data outputs from each of the demodulation channel digital data outputs of the demodulation function of the VFCT terminal shall be determined. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested with the composite VFCT baseband audio output of the VFCT terminal.

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4.6.3.3.5 Modulation function and demodulation function crosstalk tests. For these tests the VFCT terminal shall be operating in the nondiversity mode and the modulation function of the VFCT terminal shall be connected back-to-back to the demodulation function of the VFCT terminal at the composite VFCT baseband audio output and audio input. The composite VFCT baseband audio input to the composite VFCT baseband audio input of the VFCT terminal shall be adjusted to a fixed average power level of 0 dBm (with respect to the 600-ohm input impedance of the VFCT terminal) for the measurements specified herein. With 16 undistorted (isochronous and telegraph distortion less than 0.1 percent) random mark and space digital data inputs appplied to the modulation channel digital data inputs of the modulation function of the VFCT terminal and equal levels of binary FSK modulated tones outputted from the modulation function and inputted to the demodulation function of the terminal, the peak and average telegraph distortion of the digital data outputs from each of the demodulation channels of the VFCT terminal shall be determined. The levels of the binary FSK tone pair of one of the modulation channels shall be attenuated by 25 dB (with respect to the binary FSK tone pair levels of the other 15 modulation channels) at the input to the composite VFCT baseband audio input of the VFCT terminal; unless otherwise approved by the procuring activity this 25 dB level attenuation shall not be accomplished by reducing the binary FSK tone pair levels for the modulation channel selected for level reduction by adjustment of the tone pair levels being outputted from the modulation function of the VFCT terminal. The peak and average telegraph distortion of the digital data output from the demodulation channel of the VFCT terminal which corresponds to the modulation channel selected for level reduction shall be determined. The binary FSK modulated tone levels of the modulation channel selected for level reduction shall be restored to the same level as the binary FSK modulated tone pair levels of the other 15 modulation channels at the composite VFCT baseband audio input of the VFCT terminal. The binary FSK modulated tone pair levels at the composite VFCT baseband audio input of the VFCT terminal of another modulation channel shall be attenuated by 25 dB and the measurements repeated. The tests shall continue until each modulation channel, in turn, has been attenuated by 25 dB at its input to the composite VFCT baseband audio input of the VFCT terminal and a measurement of the associated demodulation channel digital data output peak and average telegraph distortion obtained. Each composite VFCT baseband audio input (A, B) of the VFCT terminal shall be independently tested with the composite VFCT baseband audio output of the VFCT terminal.

4.6.4 Operating tests. The VFCT terminal shall be subjected to operating tests to verify that the VFCT terminal will operate as specified herein. The tests shall include a check of all controls, indicators, displays, terminal input and output interfaces, circuit functions, test provisions, and adjustments. A back-to-back configuration (modulator to demodulator) of the VFCT terminal (using the VFCT terminal's modulation and demodulation functions for the modulation and demodulation functional mode VFCT terminal and the modulation function of a modulation and demodulation functional mode VFCT terminal for the demodulation functional mode VFCT terminal) shall be used to conduct a BER test with no external interference to determine conformance to the requirements of 3.4.3.3.6 or 3.4.4.1 as appropriate to the VFCT terminal functional mode classification.

4.6.5. Environmental tests. Environmental tests shall be as specified in 4.6.5.1 through 4.6.5.7.

4.6.5.1 Low temperature test. The VFCT terminal shall be subjected to the low temperature test of MIL-STD-810, Method 502.2, Procedure I and Procedure II, except as specified in a and b:

a. Procedure I:
   1. Step 1: The storage configuration shall be the unprotected VFCT terminal (without the shipping container and associated packing and packaging materials)
   2. Step 2: The chamber air temperature shall be adjusted to -62°C
   3. Step 3: The storage temperature shall be maintained for a minimum continuous period of 72 hours
   4. Step 5: Proceed to II-3.2

b. Procedure II:
   1. Step 1: The low operating temperature of the test item shall be 0°C
   2. Step 6: Proceed to step 7

The rate of temperature change shall be not less than 1°C per minute. Visual examination of the interior and exterior of the VFCT terminal shall be conducted in accordance with the Visual examination paragraph of MIL-E-16400. Operational checkout of the VFCT terminal shall include tests to establish BER in the operational modes. Failure criteria shall be in accordance with the Failure criteria paragraph of MIL-E-16400, except as otherwise specified herein. The VFCT terminal shall suffer no damage, degradation in performance, nor failures as a result of exposure to minimum storage temperature and as a result of exposure to and during operation at minimum operating temperature.

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4.6.5.2 High temperature test. The VFCT terminal shall be subjected to the high temperature test of MIL-STD-810, Method 501.2, Procedure I and Procedure II, except as specified in a and b:

a. Procedure I:
1. Step 2: The storage configuration shall be the unprotected VFCT terminal (without the shipping container and associated packing and packaging materials).
2. Step 3: The initial test conditions shall be those specified in the High temperature cycles, climatic category - Hot table of Method 501.2.
3. Step 4: The test duration shall be seven cycles of the cycle specified in the High temperature cycles, climatic category - Hot table of Method 501.2. Relative humidity control is not required.

b. Procedure II:
1. Step 2: Proceed to step 12
2. Step 12: The initial test conditions shall be those specified in the High temperature cycles, climatic category - Hot table of Method 501.2.
3. Step 13: The test duration shall be three cycles of the cycle specified in the High temperature cycles, climatic category - Hot table of Method 501.2. Relative humidity control is not required.
4. Step 14: The VFCT terminal shall be operated in a mode to establish BER for all test periods during which the ambient temperature is less than or equal to 50°C.

Visual examination of the interior and exterior of the VFCT terminal shall be conducted in accordance with the Visual examination paragraph of MIL-E-16400. Operational checkout of the VFCT terminal shall include tests to establish BER in the operational modes. Failure criteria shall be in accordance with the Failure criteria paragraph of MIL-E-16400, except as otherwise specified herein. The VFCT terminal shall suffer no damage, degradation in performance, nor failures during and after exposure to the temperature cycles of the high temperature test.

4.6.5.3 Humidity test. The VFCT terminal shall be subjected to the humidity test of MIL-STD-810, Method 507.2, Procedure III, except as specified in a through c:

a. The test item configuration shall be the unprotected VFCT terminal (without the shipping container and associated packing and packaging materials) in the operational mode.
b. Visual examination of the interior and exterior of the VFCT terminal shall be conducted in accordance with the Visual examination paragraph of MIL-E-16400.
c. Operation of the VFCT terminal during the fifth and tenth test cycles shall include tests to establish BER in the operational mode.

Failure criteria shall be in accordance with the Failure criteria paragraph of MIL-E-16400, except as otherwise specified herein. The VFCT terminal shall suffer no damage, degradation in performance, nor failures during the test and after exposure to the humidity of the test.

4.6.5.4 Salt fog test. The salt fog test specified in MIL-STD-810, Method 509.2, Procedure I, shall be applied to the VFCT terminal finishes and coatings on parts and frames, and enclosure structures as finally assembled for use. Sample corner structures and other critical sections may be used for the test. The test shall not be applied to the complete VFCT terminal. Failure criteria shall be in accordance with the Failure criteria paragraph of MIL-E-16400.

4.6.5.5 Shock test. The VFCT terminal equipments shall be shock tested in accordance with the Grade A, Type A, Class I shock tests of MIL-S-901. Each VFCT terminal equipment of the VFCT terminal configuration shall be individually shock tested using the lightweight test of MIL-S-901. The equipment shall be hard-mounted to the test fixture which shall be hard-mounted to the test machine. The mounting of the equipment to the test fixture shall be representative of that used for shipboard installation and the mounting shall be as approved by the procuring activity. Simulated loads, such as weights or other dummy masses, shall not be used. The test shall be performed with the normal assemblies and units in their functionally operating positions. The VFCT terminal shall be operated before, during, and after the shock test. Visual examination of the interior and exterior of the VFCT terminal equipment shall be conducted in accordance with the Visual examination paragraph of MIL-E-16400 prior to and at the conclusion of the shock test of the equipment. The VFCT terminal equipment shall be operated in a mode to determine bit errors and BER prior to, during, and after application of the hammer blows of the shock test. The VFCT terminal equipment shall suffer no degradation in performance nor failures as a result of exposure to the hammer blows of the shock test.

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4.6.5.6 Vibration test. The VFCT terminal equipment shall be subjected to the Type I vibration test of MIL-STD-167-1. Each VFCT terminal equipment of the VFCT terminal configuration shall be individually tested or the VFCT terminal configuration shall be tested as a whole. The upper vibration frequency limit shall be 50 Hz. The VFCT terminal equipment(s) shall be hard-mounted to the test machine; the method of mounting shall be subject to procuring activity approval. Simulated loads, such as weights or other dummy masses, shall not be used. The test shall be performed with the normal assemblies and units in their functionally operating positions. The VFCT terminal shall be operated before, during, and after the vibration test. Visual examination of the interior and exterior of the VFCT terminal equipment shall be conducted in accordance with the Visual examination paragraph of MIL-E-16400 prior to and after conclusion of the vibration test. The VFCT terminal shall be operated in a mode to determine BER prior to, during, and after conclusion of the vibration test. Failure criteria shall be in accordance with the Failure criteria paragraph of MIL-E-16400, except as otherwise specified herein. The VFCT terminal shall suffer no damage, degradation in performance, nor failures during and after exposure to the vibration test.

4.6.5.7 Inclination test. The VFCT terminal shall be subjected to the inclination tests of the Inclination test limits (except submarine installations) paragraph of MIL-E-16400, except as otherwise specified herein. The VFCT terminal shall be energized and fully operating (except as otherwise specified herein) during the test. Visual examination of the interior and exterior of the VFCT terminal shall be conducted in accordance with the Visual examination paragraph of MIL-E-16400 prior to and after the inclination test. Performance tests, including tests to establish BER in the operational modes, shall be conducted on the VFCT terminal prior to, during, and after the inclination test. Failure criteria shall be in accordance with the Failure criteria paragraph of MIL-E-16400, except as otherwise specified herein. The VFCT terminal shall suffer no damage, degradation in performance, nor failures during and after exposure to the inclination test.

4.6.6 Interchangeability demonstration. The VFCT terminal shall be examined and tested to verify and demonstrate interchangeability by interchanging assembly, subassembly (for example, printed circuit cards), and modules between two VFCT terminals of like functional mode, in-band frequency diversity, and resident channel capability. Interchangeability of piece parts shall be verified by examination. Each of the VFCT terminals shall operate in accordance with the requirements of this specification prior to and after interchange.

4.6.7 Electrical tests. The VFCT terminal shall be subjected to the steady-state voltage and frequency, transient voltage, transient frequency, power interruption, and power and power factor tests of the Electrical tests paragraph of MIL-E-16400, except as otherwise specified herein. For each test a back-to-back (modulator to demodulator) configuration of the VFCT terminal (using the VFCT terminal's modulation and demodulation functions for the modulation and demodulation functional mode VFCT terminal and the modulation function of a modulation and demodulation functional mode VFCT terminal for the demodulation functional mode VFCT terminal) shall be tested. A BER tester shall be used to determine bit errors during the test.

4.6.7.1 Steady-state voltage and frequency test. For this test, the VFCT terminal normal voltage shall be 115 VAC, single-phase. The VFCT terminal normal frequency shall be, in turn, 50 Hz and 60 Hz. Capability of the VFCT terminal to operate in the operating modes from AC source power with the specified voltages and frequencies within the power factor limitation of DoD-STD-1399, Section 300, shall be determined. BER shall conform to the requirements of 3.4.3.3.6 or 3.4.4.1 as appropriate to the VFCT terminal functional mode classification.

4.6.7.2 Transient voltage test. For this test, the VFCT terminal steady-state voltage shall be 115 VAC, single-phase, and the steady-state voltage frequency shall be 60 Hz. The VFCT terminal shall operate without bit error, fault, and interruption before, during, and after the voltage transient.

4.6.7.3 Transient frequency test. For this test the VFCT terminal normal frequency shall be, in turn, 50 Hz and 60 Hz, and the steady-state voltage shall be 115 VAC, single-phase. The VFCT terminal shall operate without bit error, fault, or interruption before, during, and after application of the frequency transient.

4.6.7.4 Power interruption test. The power interruption test shall be conducted, in turn, for AC source power inputs at frequencies of 50 Hz and 60 Hz; the steady-state voltage shall be 115 VAC, single-phase. An additional power interruption test for a power interruption of 120 seconds shall be conducted. The VFCT terminal shall operate without error before and after interruption of AC source power. The VFCT terminal shall resume normal operation without operator intervention after reapplication of AC source power.

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4.6.7.5 Power and power factor test. For this test, the VFCT terminal normal line voltage shall be 115 VAC, single-phase, and the normal line frequency shall be, in turn, 50 Hz and 60 Hz.

4.6.8 EMC test. The VFCT terminal shall be tested in accordance with tests specified in MIL-STD-462 to determine conformance to the radiated emissions and conducted emissions requirements specified in 3.10.

4.6.9 Enclosure test. The enclosures of the equipments and units of the VFCT terminal shall be subjected to the dripproof test (15 degrees) of MIL-STD-108. Pretest and post-test of the VFCT terminal shall include, as a minimum:

a. Visual examination (interior and exterior) in accordance with the Visual examination paragraph of MIL-E-16400
b. Operational and functional tests of controls, indicators, and displays
c. Performance tests, including tests to establish BER in the operational modes

Failure criteria shall be in accordance with the Failure criteria paragraph of MIL-E-16400, except as otherwise specified herein. The VFCT terminal shall suffer no damage, degradation in performance, or failures after exposure to the dripproof (15 degrees) enclosure test.

4.6.10 Airborne noise test. The airborne noise test shall be applicable only to VFCT terminals employing forced air cooling. For this test the VFCT terminal shall be tested to determine compliance with the airborne noise requirements of 3.12.3. The VFCT terminal shall be installed on a hard, sound-reflecting plane at the center of an imaginary hemisphere having a 6-foot radius. Sound pressure readings in the form of an octave band analysis, using an octave band filter and a precision sound level meter, shall be obtained at 12 uniformly distributed points on the surface of the hemisphere. The octave band analysis data shall then be used to compute sound power in accordance with MIL-STD-740. The VFCT terminal shall be operated in the mode in which the VFCT terminal airborne noise level is the greatest.

4.6.11 Structureborne noise test. The structureborne noise test shall be applicable only to VFCT terminals employing forced air cooling. For this test the VFCT terminal shall be tested to determine compliance with the structureborne noise requirements of 3.12.3. The VFCT terminal shall be mounted resiliently (even though this may not be the case for normal installation) and vibration accelerometers shall be used to record broadband and one-third octave band measurements. The test shall be performed in accordance with the procedures for Type 3 equipment of MIL-STD-740. The VFCT terminal shall be operated in the mode in which the VFCT terminal structureborne noise level is the greatest.

4.6.12. Safety verification. Examinations, analyses, and tests shall be conducted to verify that the VFCT terminal conforms to the safety requirements specified in 3.14.

4.6.12.1 Leakage current test. Leakage current of each equipment of the VFCT terminal having AC power applied shall be measured with 115 VAC, single-phase source power applied to VFCT terminal equipment at source power line frequencies of, in turn, 50 Hz and 60 Hz. The leakage current of each equipment of the VFCT terminal having AC power applied (with any units deriving power therefrom connected thereto) shall be individually determined; other units of the VFCT terminal having AC power applied shall not be connected to the unit under test.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE VFCT TERMINAL EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

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4.6.12.2 VFCT terminal test connections. After AC power removal, each VFCT terminal equipment directly connected to an external AC power source and VFCT terminal units deriving power from that equipment shall be placed, in turn, on an insulated surface. All safety ground conductors between the VFCT terminal equipment and VFCT terminal units deriving power from that equipment shall be intact. The safety ground conductor between the VFCT terminal equipment and the AC source power shall be opened during the test. OBSERVE WARNING STATEMENT. The VFCT terminal equipment shall be connected as shown in FIGURE 5 or single-phase AC source power.

4.6.12.3 Measurement. Leakage current shall be measured on the VFCT terminal equipment (and on VFCT terminal units deriving power from the equipment) in its normal operating configuration. VFCT terminal equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor, when equal to 7.5 volts, represents 5 mA of leakage current. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housing, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part to ground for every combination of switch positions available in FIGURE 5. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.6.13 Software program performance tests. If the VFCT terminal contains one or more microprocessors, the software (firmware) of each processor of the VFCT termial shall be subjected to the stress tests of the Software acceptance paragraph of DoD-STD-1679. Software acceptance test duration for each processor shall be a minimum of 25 continuous hours.

4.7 Reliability design qualification tests. Each of two configurations of the VFCT terminal shall be subjected to a reliability design qualification test in accordance with Test Plan XXC of MIL-STD-781, except as otherwise specified herein. The two VFCT terminal test configurations shall be the VFCT terminals specified as A and B in Table IX. A minimum of two and a maximum of three VFCT terminals of each test configuration shall be tested. Reliability design qualification testing shall be in accordance with the Reliability test procedures paragraph of MIL-STD-781 for Category 3A equipment, except that the combined environments for shipboard equipment shall be as specified in a through e:

a. Vibration stress (continuous). During the test period, the VFCT terminal shall be subjected to continuous vibration (24 hours per day) in accordance with the requirements of the Vibration stress (continuous) paragraph of Appendix B to MIL-STD-781 except that the upper swept frequency shall be 50 Hz.

b. Temperature cycling. The VFCT terminal shall be subjected to continuous temperature cycling as specified herein during the test. Each 24 hour period shall be broken up into three 8-hour temperature cycles each of which shall be identical. During each temperature cycle, the ambient temperature of the VFCT terminal shall be cycled from 0°C to 50°C to 0°C. During each temperature cycle, the continuous dwell time at 0°C shall be a minimum of 3 hours and 45 minutes; during each cycle, the continuous dwell time at 50°C shall be a minimum of 3 hours and 45 minutes. The rate of chamber temperature change shall be not greater than 5°C per minute.

c. AC input voltage variation. The VFCT terminal shall be subjected to the AC source power input voltage variations as specified herein during the test. Each 24-hour period shall be divided into three 8-hour time periods. During the first 8-hour period, the VFCT terminal shall be operated from an AC source power voltage that is 7 percent higher than the nominal AC source voltage. During the second 8-hour period, the VFCT terminal shall be operated from an AC source power voltage that is equal to the nominal AC source power voltage. During the last (or third) 8-hour period, the VFCT terminal shall be operated from an AC source power voltage that is 7 percent lower than the nominal AC source power voltage. The nominal AC source voltage shall be 115 VAC rms, single-phase. The nominal AC source voltage frequency shall be 60 Hz.

d. VFCT terminal operation. The VFCT terminal shall be operated continuously during the entire period of the test, except as otherwise specified herein (see e). During operation, the performance of the VFCT terminal (including number of data errors generated and the time period during the test cycle at which they occurred) shall be continuously monitored.

e. AC source power interruption. In every 24-hour test period during the operation of the VFCT terminal in its operational mode, the AC source input power to each unit of the VFCT terminal shall be simultaneously interrupted for a period of 120 seconds (+0,-5 seconds) and the capability of the VFCT terminal to continue operation without operator intervention determined. AC source power interruptions shall be performed during dwell periods of 0°C and 50°C alternately during the 24-hour test periods.

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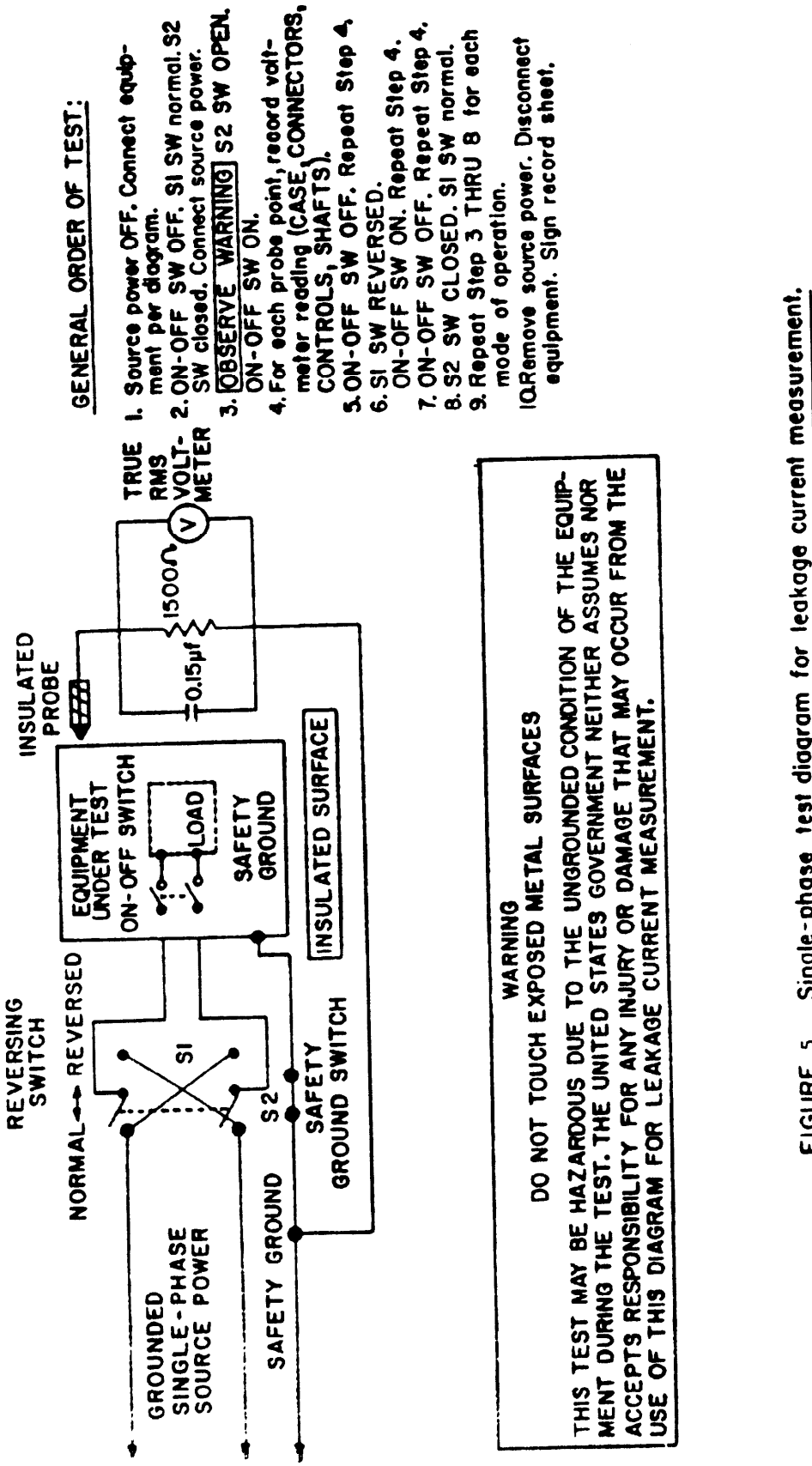


FIGURE 5. Single-phase test diagram for leakage current measurement.

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The procuring activity shall be immediately notified of every failure occurring during reliability design qualification testing. The final decision concerning relevancy or nonrelevancy of each failure shall be made by the procuring activity using MIL-STD-781 as a guide. A failure initially classified as relevant may be subsequently reclassified as nonrelevant only if a failure analysis clearly indicates that the failure was classified improperly and written approval of the procuring activity is obtained. Only relevant failures will be considered in making an accept or reject decision, but all failures shall be subject to failure analysis by the contractor. If the equipment is rejected as a result of reliability testing, the matter shall be referred to the procuring activity for direction and the contractor shall immediately notify the procuring activity of the cause(s) of such rejection and provide a proposed plan for correction of the deficiencies.

4.8. Maintainability demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the corrective maintenance time requirements are fulfilled.

4.8.1 Demonstration conditions. Conformance to the VFCT terminal maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of assemblies, subassemblies, printed circuit cards and individual mounted chassis piece parts at the organizational level. Not less than 100 candidate corrective maintenance tasks shall be identified (selected in accordance with APPENDIX A of MIL-STD-471). The information specified in a through c shall be made available for each candidate task:

a. Description of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 50 tasks for the demonstration.

4.8.2 Accept or reject criteria. The accept or reject criteria for the demonstration of the corrective maintenance times shall be as specified in TABLE XI. An accept decision is reached for MTTR when the number of observed corrective maintenance task times which exceed the required value of 45 minutes is equal to or less than that shown in TABLE XI. No maintenance times are permitted to exceed the M_{maxct} of 60 minutes.

TABLE XI. Corrective maintenance acceptance criteria.

	MTTR	M_{maxct}
Confidence level	90 percent	90 percent
Acceptance level (allowable number of tasks with task times greater than column heading requirement)	20	0
Task sample size	50	50

MTTR calculated from task times for task sample size shall be less than or equal to MTTR requirement.

4.8.3 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the VFCT terminal will be installed.

4.8.4 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

4.8.5 Technical documentation. Technical documentation used in the demonstration shall be limited to the technical manual and related maintenance documentation delivered with the VFCT terminal.

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4.8.6 Rejection. Failure to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the matter shall be referred to the procuring activity for direction and the contractor shall immediately notify the procuring activity of the causes of such rejection and provide a proposed plan for correction of all deficiencies.

4.9 Workmanship. Workmanship shall be specified in 4.9.1 through 4.9.2.2.

4.9.1 General workmanship. The VFCT terminal, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of 3.19.1. Each solder connection and associated wiring or leads shall be visually examined.

4.9.2 Workmanship screen. Each VFCT terminal shall withstand a defect detection vibration screen and temperature cycling as specified in 4.9.2.1 and 4.9.2.2.

4.9.2.1 Random vibration. Prior to conducting temperature cycling (see 4.9.2.2), random vibration testing shall be performed on each VFCT terminal at the end item level. All the hardware, including cables and connectors, shall be exposed to random vibration testing. The vibration shall be random or, subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 6. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz, and 100 Hz for vibration frequencies above 500 Hz, shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks shall be limited to three times the rms acceleration level. The VFCT terminal shall be energized during vibration and digital input and audio input signals applied to observe any abnormal conditions of the audio output and digital output functional characteristics. All failures occuring during screening shall be corrected and the vibration resumed for an additional continuous 10-minute test period without failure.

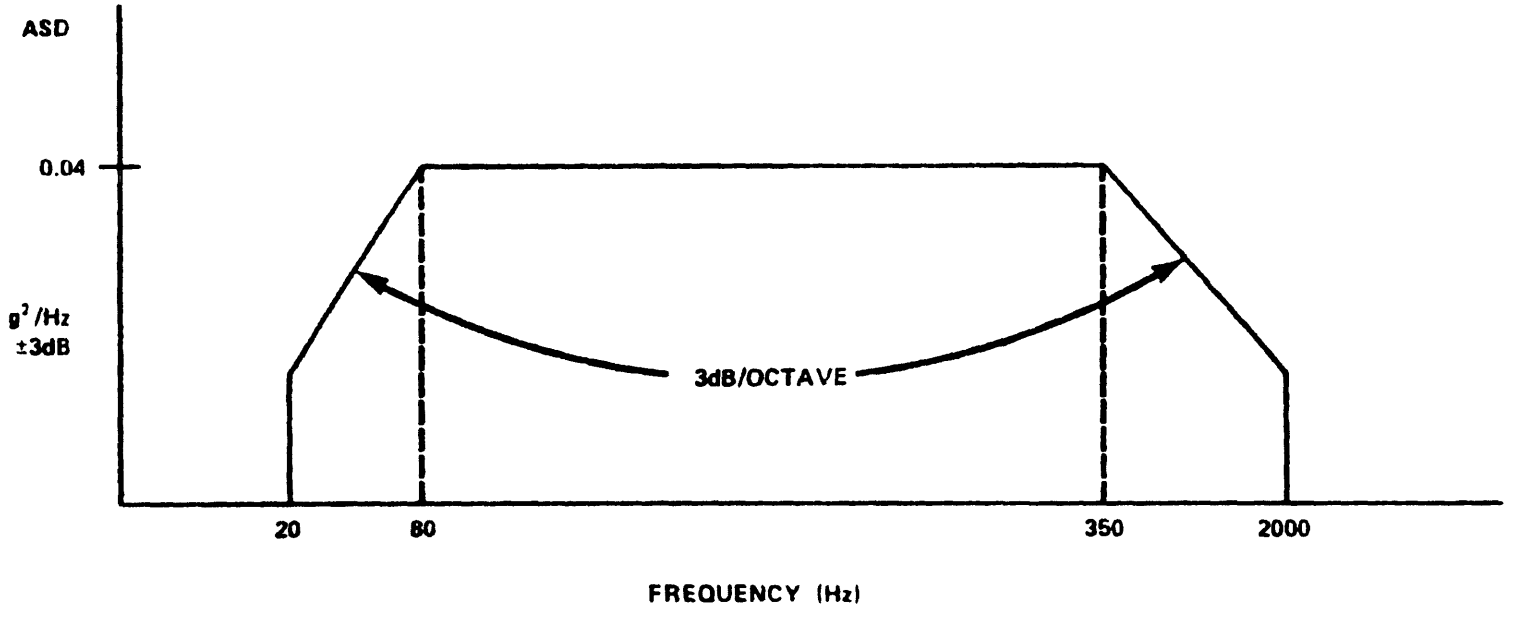


FIGURE 6. Random vibration curve.

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4.9.2.2 Temperature cycling screen. Each VFCT terminal shall be subjected to 10 cycles of the temperature curve shown in FIGURE 7. The temperature rate of change shall be not less than 5°C per minute. AC source power to the VFCT terminal shall be turned on and off at the indicated times. The VFCT terminal shall be positioned for maximum exposure to the change in temperature. Where performance measurements are called for in FIGURE 7, as a minimum, a functional operating test involving determination of either data BER or number of data errors shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the VFCT terminal reaches 80 percent of the chamber temperature. When failures occur, the VFCT terminal shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.10 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

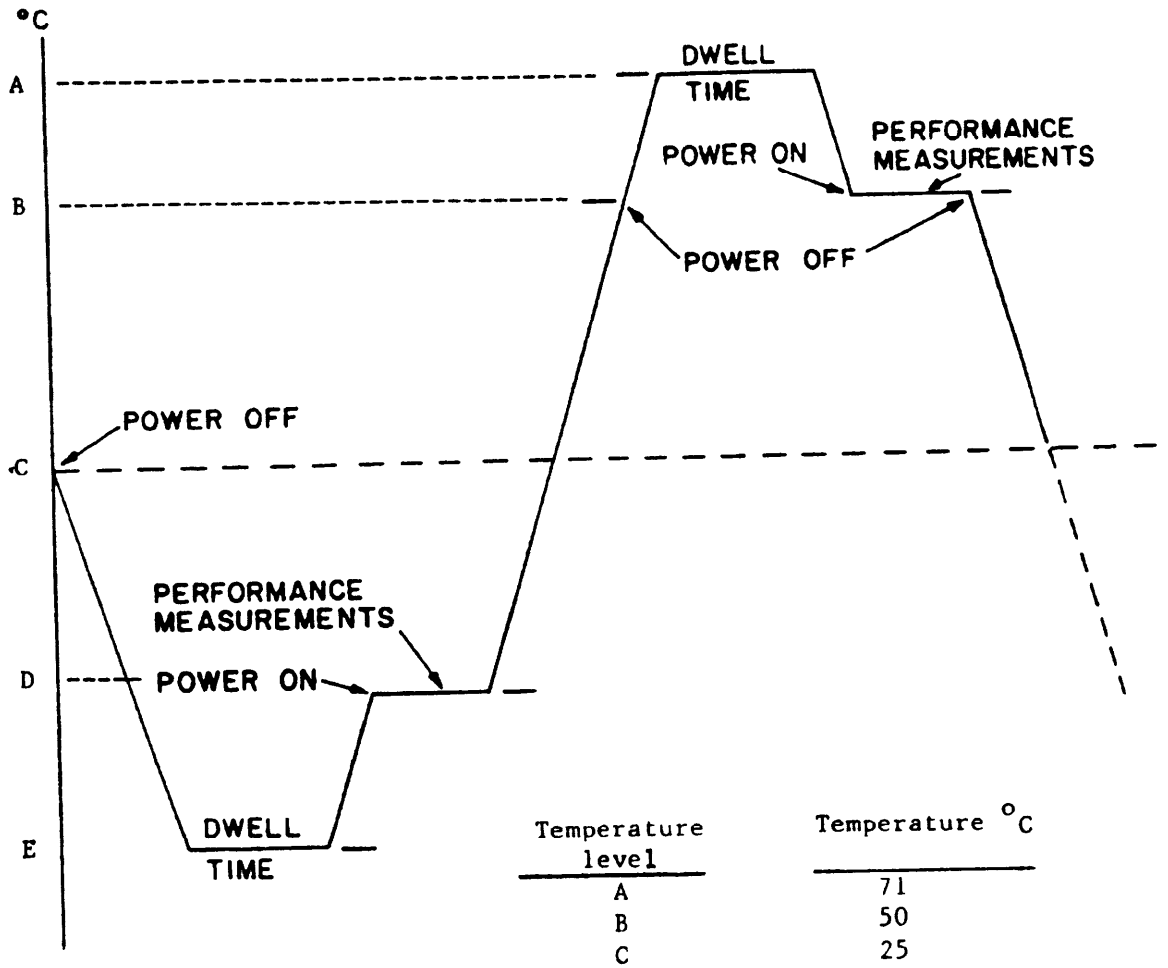
5.2 Packaging, handling, transportation, and storage (PHTS). PHTS requirements shall be considered in design and preparation for delivery of equipment, accessories, and parts to facilitate handling, movement, and storage, and to minimize damage therein. The VFCT terminal and components shall be limited in size and weight as specified in this specification. Units identified as manportable will be provided with lifting or carrying devices which enable the equipment to be easily transported by operator and maintenance personnel. Special consideration shall be given to the combat environment and environmental extremes, specified in the specification, in which the system will be deployed, operated, and stored.

5.3 Packaging of electrostatic discharge sensitive (ESDS) items. ESDS items shall be packaged in ESD protective packaging, labeled with ESD markings and cautions. ESDS items shall have their leads and connector terminals shorted with noncorrosive ESD protective conductive foam, shorting clips, or bars. ESDS items shall be protected to prevent physical damage and to maintain leads and terminals in the manufactured condition under handling and transportation environments. Material used for this protection shall be noncorrosive and provide protection against triboelectric generation of electrostatic potentials. Conductive outer wrapping shall be used to protect ESDS items from damage caused by electrostatic fields, triboelectric charging, and direct contact with charged objects. The outer wrap shall be marked with the MIL-STD-129 sensitive electronic device caution label and the caution statement:

CAUTION

OBSERVE PRECAUTIONS FOR HANDLING
ELECTROSTATIC SENSITIVE ITEMS

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Temperature level	Temperature °C
A	71
B	50
C	25
D	0
E	-62

FIGURE 7. One cycle of temperature curve (not to scale).

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6. NOTES

6.1 Intended use. The VFCT terminals covered by this specification are intended for use in line-of-sight, extended line-of-sight, and beyond line-of-sight HF radio communications circuits between ship and shore. The VFCT terminals will be used as shipboard DCEs for the modulation and demodulation of multichannel (up to and including 16 channels) frequency division multiplex binary frequency shift keyed communications signals. The digital interfaces for the VFCT terminal will be with cryptographic equipments capable of operation at isochronous data signaling rates of 75 bps per channel. The digital data inputs of the VFCT terminal will also interface with 75 bps isochronous digital data having a total peak distortion of up to and including 35 percent that was derived from demodulation of single channel binary FSK signals by modems such as: a) CY-2460( )/SGC and b) AN/URA-17( ). Asynchronous operation and operation at data signalling rates of other than 75 bps per channel will not be required. The timing of the modulation channel digital inputs to the modulator function of the VFCT terminal and the demodulation channel binary FSK audio tone inputs to the demodulator function of the VFCT terminal will be such that the bit transition (mark to space, space to mark) for each of the channels will not necessarily be coincident in time with the bit transitions of the other channels. The audio interfaces for the VFCT terminal will be with shipboard radio sets, transmitters, and receivers such as: a) AN/URT-23( ), R-1051( )/URR; b) T-1322, R-1903/URR; and c) AN/URC-81( ). The VFCT terminal will interoperate with the following VFCT terminals that will be installed in the communication system at the other end of the ship-to-shore HF communications circuit: a) AN/FCC-66; b) AN/FCC-67; c) AN/FCC-68; d) AN/FCC-69; e) AN/FCC-70; f) AN/FCC-71; g) AN/FCC-72; h) AN/USQ-83(V) (VFCT mode); i) AN/UCC-1(V); j) AN/UCC-1A(V); k) AN/UCC-1B(V); l) AN/UCC-1C(V); m) AN/UCC-1D(V).

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Functional mode capability (see 1.2.1)
c. Modulation function in-band frequency diversity capability (if modulation and demodulation functional mode capability is specified (see 1.2.1 and 1.2.2.1))
d. Demodulation function in-band frequency diversity capability (see 1.2.2.2)
e. Modulation function resident channel capability (if modulation and demodulation functional mode capability is specified (see 1.2.1 and 1.2.3.1))
f. Demodulation function resident channel capability (see 1.2.3.2)
g. Quantities of VFCT terminals and classification of each (see 3.3 and 3.4)
h. Number of first article samples to be submitted (see 4.3)
i. When reinspected production control and environmental sample units may not be accepted (see 4.4.4)
j. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article inspection is required, the item(s) should be first production units. The first article should consist of the number of units specified in the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-29500	TERMINAL, VOICE FREQUENCY CARRIER TELEGRAPH (VFCT), AN/UCC-1( )(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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